

3rd Applied Active Inference Symposium ~ "Enacting Ecosystems of Shared Intelligence" ~ 2nd session

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hello and welcome everyone welcome back this is the second interval of the third applied active inference Symposium at the Active inference Institute on August 22nd 2023 this is going to be another pact and exciting interval and we're kicking it off with Jean Francois katir a collective of theorizers First Steps so JF thank you for joining and to you for this presentation and if anybody has questions for this presentation or any of the others please just put it in the live chat and I'll do it I can so thanks JF to you well uh thank you Daniel um I'm a software engineer a company called smart rent where I write software for smart home systems and I also work on a research project at the Active inference Institute called the Robotics and embodied project my current focus is on supervised learning in active inference agents in my presentation today which is titled a collective of theorizers First Steps um I'll first start with a brief recap of the project then dive into recent progress and then I'll conclude with what I see as the next steps of the project well since 2017 I've been experimenting with Lego robots and models of cognition I do this for my own education because like I think most of us I understand something better when I build it what you see here is the my latest robot model it's a Rover it has a whole bunch of sensors and a number of effectors actuators as a matter of fact Those sensors and actuators can be understood as forming the marker of blanket of my robot last year I presented at the Symposium about the history of the project it's uh then status and it's ambitions I presented a cognitive model that implemented predictive processing from an active inference perspective so they were uh genetic models there was predictions there was prediction errors and um the these generative models animated the robots so that they would roam avoid obstacles uh observe their uh companion and also in a very simplistic way build a theory of Mind of the other robot they could uh from observations infer that uh the other robot had detective food food being presented here as a sheet of paper on the floor and then would uh kind of try to track the other robot to also get to the food and and fun stuff like this the

implementation uh combined uh multi-processing and functional Computing there was nothing probabilistic in the implementation I thought maybe you would uh see them in action that's what's presented last year but just to get a sense of what they do I have two robots here um one named Carl and Andy I named them before I knew I was going to present at the conference um so uh they benefited from a number of training sessions they uh learned how to um uh find which policies would achieve their goals better than others and uh one here on the left found the food rather right away but got closer to the pedestal uh where the beacon is that simulates the scent of food um fear was getting into a collision and then backs off in a hurry as we'll see the other robot observes all this sees the other the the robot backing off as being in a panic and uh decides to share in in this emotion and backs up as well so you know uh fun stuff the um since from the beginning I've implemented different cognition models but they all have in common the fact that they Implement a Society of Mind um so what is a Society of Mind a Society of Mind is a concept by which um the uh the mind is not a monolithic uh structure but a composition of simple actors independent actors that interact with each other in simple ways that uh view of the Mind was put forth by by Marvin Minsky um 50 years ago so this is not recent again what I presented last year was a Society of Mind containing a hierarchy of I call them cognition actors each cognition actor is an independent process each one has a scope a level of abstraction as well so for example you would have a commission actor that's concerned with the location of food and it would have beliefs and perceptions about food location and these would feed into a higher level cognition actor let's say food approach which is concerned about getting closer to the food and so forth and so on these connection actors communicate again in simple ways this is the Society of Mind and and they communicate by emitting predictions predictions about the beliefs of other cognizant actors and they communicate by emitting prediction errors when predictions made about their beliefs by others uh are inaccurate that leads to uh coaching actors processing these uh prediction errors and combining them with their own predictions to create a an updated set of perceptions that are synthesized into beliefs and these beliefs lead to um actions in order to eliminate negative beliefs and Val or validate positive beliefs now it's from the interactions of all these connection actors that seemingly purposeful behaviors emerge so that was successful but learning was very limited this um this hierarchy of congression and actors uh was given was pre-defined and the only learning that a robot would would do was to discover which action policies tended to be more effective so clearly my robots are not monolithic um active inference agents um so the question is why would I use a Society of Mind architecture well because I I

subscribe uh to the uh notion that all intelligence is collective intelligence and and this paper makes uh the argument uh quite cogently and I'm gonna I'm gonna cite a number of papers which were important to the evolution of my thinking this is one of them this paper sees intelligence as a and not a property it's a process enacted by the interacting parts as opposed to again a property of individuals so a Society of Mind is basically that it's a it's actually a um a number of interacting processes um that together from which emerge apparently intelligent behaviors now my my own personal current definition of intelligence is uh self-sustaining and active sense making since making is really important by autonomous agent system in a dissipative and dynamic environment and this guides um all the work I do in this project now if you want a um description a detailed description of uh where my project stood at the a year ago there's a paper that was published that I published on xenodone which you can look up now over the last year I wanted to move away from a pre-built a given Society of Mind and toward a learned Society of Mind I wanted to see if um I could program an autonomous robot to evolve in Society of Mind through its interactions with its environment now one might think that programming autonomy is an oxymoron um it's it's it's it's it would be a good point but I don't think it is um if if the program I write and and install my robots uh imparts uh constitutive autonomy which is uh enable the robot to constitute its own identity and if it imparts adaptivity which uh enable the robot to uh modify itself from its interactions with the environment then I think that um uh the the robot uh will be truly autonomous now for autonomy to um to exist inherently in the robot something in the robot must be at stake and in it's different it's in this case it's the survival of the robots Society of mind now the physical structure of the robot is not uh at stake it's survival is not at stake unless of course it falls off a shelf but as I intend to um have my robot develop its Society of Mind from experiences I also expect that if it fails that the Society of mine will perish in its attempt to to grow and sustain itself so that's what's at stake in in this current and going forward in this project is the robots survive the survival of the robot Society of mine and and this survival would be an expression of being by doing uh the robot need will need to act an interact in its environment in order to uh survive now whatever sense it's going to make of its environment will then be grounded in this survival imperative in essence the robot Society of mine will have skin in the game if this weren't the case then sense making would actually reside somewhere else it would reside in in in the mind of the of the programmer in my mind as I observe uh the robots but I would like the sense making to be grounded in the survival of the robot's mind itself so that's key now what do I mean by an evolving and growing uh Society of Mind so instead of the given

Society of Mind which I I showed uh earlier I want um I want the cognition actors to uh be created to connect with each other dynamically through interactions with the environment um so I want an active uh self-organizing self-optimizing Collective of cognition actors now is this feasible well that's a big question um in order to and in trying to answer this question uh I'll be um also answering uh questions like what must be given a priori and what can be discovered well I already have elements of an answer I know that the um the sensors and the uh effectors and the um the cognition the Primitive cognition actors that wrap them will be given this this will be like what you're born with basically and there might be a um a meta cognition actor which uh role will be to uh oversee and guide the evolution of uh all other cognition actors but that's that's my hypothesis I can imagine a test environment for my robot that will test its ability to survive I can imagine for example that um as the the robot uh moves around or computes it it uses a Limited store of energy it's simulated and that store of energy is replenished when the robot consumes food that would be by being on top of pieces of paper colored pieces of paper on the floor um and I'm going to make sure that it needs two sources of food in order to survive right that would be represented by a yellow paper and a green paper for example so that to prevent the the robot from just simply finding one one source of food and just stationing itself over it so the environment will also contain obstacles so the robot will need to learn how to navigate avoid obstacles locate different sources of food get to them and make sure that it it alternate between various sources of food in order to survive and the um Society of Mind that will evolve will uh hopefully evolve to successfully do this else if it doesn't then it will um shrink as resources uh disappear and uh essentially uh die so that's when it's at stake uh for uh this robot now this effort employs a a number of Frameworks and by framework I mean a useful system of concept and constraints that that guide uh the implementation well there's obviously the the free energy principle and the active inference framework however I see this the IC active inference as and as if framework it describes the what what must be achieved in this case uh reduction of the agent's uh variational free energy but it doesn't guide me as to the implementation how to build the robot for this um I need as is Frameworks and um I'll be using uh two Frameworks one which I've been using since uh the beginning of the project which is uh the actor model uh the actor model uh views computation as um uh diversity of processes processes that are independent have their own private internal State and who communicate with one another strictly through messages um yesterday in uh interval one uh Keith Duggar uh presented on the actor model and made the case that we should use the actor model to implement active inference agents well I wholeheartedly agree

with them the other uh framework that I'm going to be using is a special case of symbolic AI called the app reception engine and much of the presentation will be about the opposition engine and its implementation so here's where we are uh is the project is located at the intersection of active inference as a domain the what and Society of Mind as a architecture and symbolic AI as a form of computing um so that's where the project is at this intersection no so where to begin so I I want to um uh dive into a more extensive form of learning and the first step uh logically is to learn how to predict so um I want to enable a single uh cognition actor we'll start with a single cognition actor to um learn how to make sense of its local environment it's so-called unvelt and making sense implies at a minimum to be able to predict incoming Sensations so it needs to learn to predict so what will be given to a cognition actor well a uh there will be a history of uh Sensations uh broken into discrete uh units of time so time n minus 3 and N minus two n minus 1 time n which is the the present moment so these would be remembered observations and then what we want to get out of this is the ability to predict the next incoming set of Sensations at time uh T equals n plus 1 and plus two and for this to be able to predict uh future observations we need some kind of predictor function that is learned from the remembered observation now this predictive capability can be built in two very general ways one is from statistics uh so doing pattern analysis and and being able to predict what's most probable which is the standard Uh current machine learning approach or we could predict from an understanding of uh the observations by developing a causal model uh of these of what uh produce these Sensations and from this understanding uh predict what rationally should be observed next so this is all about sense making and now what is sense making how do I understand sense making well to rationally predict incoming sensory inputs uh one must make sense of them that's what making sense means to me um and to make sense of sensory inputs means to derive meaningful experiences from them it's not just data it's not just pieces of data it's a there must be meaningful experiences and by experience uh I mean a conceptualization of the sensations and a unification of them in time in space so making sense of these inputs will mean to produce meaningful experiences that are conceptualizations and unifications of the sensations now an experience is Meaningful if it is underwritten by a causal model so the expense is uh is uh is perceived as the consequences of a latent generative model a generative process that we have modeled and I want meaning to be inherent to the agent and that only happens if the agent is truly autonomous and if this meaning is grounded in the survival imperative as uh discussed earlier so how does expanding work how can that be put into computer code surprisingly for this we refer to the philosophy of Emmanuel Kant a

reverse engineering approach um asking himself what must entities do to achieve experience this is akin to the free energy principles high road which can be paraphrased as what most organisms do to maintain their existence so in manual con uh we've tried to reverse engineer um uh cognition um asking himself what's the minimal cognitive apparatus needed by an entity to have experiences and and that he he documented in his critique of pure reason just a little parenthesis um the the meaning of of the title critic of pure reason is not what I thought it was it's actually translates more closely to the case for a period recognition a critique is is a legal term is where you make your case and pure reason uh we translate nowadays is a priori cognition his work wanted to establish what must happen to create an experience that's coherent that is Unified in time and space um and to reverse engineering cognition as a system that is both complete and essential that is minimal okay so let's let's um I'm gonna try to give you the uh posted uh postage stamp uh version of uh Emmanuel Theory uh focusing on the synthetic Unity of Apperception well first of all there's the real world which is outside of our direct experience uh it's the numina it's forever hidden from us um as an as is uh reality but we do uh it it impinges on our sensorium and then so we have a number of intuitions site sound touch smell um that are initially uh separate and and then we need to uh Network them connect them both in time and in space in space um uh is is sound and and and and the site uh describing a single thing um uh is one thing uh behind or uh inside another um and in time is this happening before after uh something else and and then at a higher level uh meaning is given to these uh networked intuitions Sensations uh via Concepts and and judgments rules uh which are generalizations as to what can and cannot be um and this is what we experience now it it turns out that a con synthetic Unity of that perception is a blueprint for automating a sense making and it's kind of uh interesting I think that 18th century philosophy would be uh relevant to uh 21st century uh technology development and this is what happened and and and was published uh by Richard Evans and and all in the paper making sense of sensory input where um they developed the app reception engine they took um synthetic Unity of our perception as software requirements and successfully implemented uh them into a piece of software the app reception engine and applied it to a number of uh exercises um and where they got a very good result so the opposition engine is in um an instance of machine learning it's unsupervised machine learning and it operates on very small data sets um and generates uh human readable uh generative models and when I read the paper I realized well that's exactly uh what my robots needed so what doesn't that perception engine do well given a a sequence of observed States it uh finds a generative model that can recreate past States but most

importantly predict uh future uh States and the state is defined as a set of simultaneous uh observations Sensations intuitions an app reception engine searches for a causal theory that can recreate observations I say searches because this causal theory is not determined by the observations it has to be found it has to be discovered but once it is found then it can be validated against the observations and see and see if it can recreate them and augment them into the future as well as into the past so what what is a causal Theory a causal theory is a logic program that has a number of components um they're in the causal Theory there will be um the objects and the predicates from The observed uh relation so from the observations we can extract what objects were observed and what properties of these objects were observed and maybe what relationships and these these objects were observed that's the start then we have latent object types objects and predicates so we may want to imagine causal Theory imagines a hidden objects maybe in types of objects and maybe hidden uh properties and relationships between uh objects uh latent meaning unobserved and given both The observed and unobserved um uh objects and and and and predicates it derives rules first of all constraints on those predicates what's permissible so for example being in front of cannot uh a cannot be in front of B N at the same time B and at the same time behind B so an object cannot be in front of another and behind it so they're constrained some predicates And Then There are rules that apply to any simultaneous sets of observations what what uh they must conform to Then There are rules that given a state will infer the next state and then maybe some initial state from which we can uh uh uh run the causal Theory what makes a causal Theory Unified well first of all it needs to be unified in order to make sense of the there are various Dimensions so if a causal Theory involves a number of objects all these objects must be directly or indirectly related there's no object that just floats in space totally independent of the other objects so they must all be related um so they're they're spatially Unified all predicates that make up the uh causal Theory like on off the hind in front they must be uh constrained so that um for example in front cannot be at the same time as behind or that a light cannot be both turned on and turned off so there's there's some restrictions on uh the predicates and that creates conceptual Unity and in the aesthetic Unity where all simultaneous relations must satisfy the static rules and temporal Unity where all the states must be sequenced by causal rules we'll see examples let's start with a an example here of a set of observations what we are observing are two lights uh and the lights can either be at any discrete Moment In Time either on or off so here we have a sequence of uh observations uh and One Moment In Time the first light was off and the second night was on then the first light was on the certain light

was off then both were on Etc and I put the grape bars there um to show that maybe the observations aren't complete so on one stage we can only see the second light or there may be other lights or other objects that we do not see but that's what we observe you feed these observations um in discrete time into the art perception engine and the opposite perception engine searches for a causal Theory that when applied to an initial condition let's say that the light A is often light B is on it will create a a trace of uh recreated observations that um cover is a superset matches the initial observations and if this happens then our causal theory is a good one now the causal Theory May infer uh the existence of hidden objects hidden relations and whatnot it may actually need to okay so here's a an example of a causal theory that is generated by my own implementation of the perception engine because I I re-implemented the the perception engine as described in the paper by Richard Evans and all and um I ran it on this the set of observations about lights on uh two lights uh one on one off at any point in time and it came up with it found a result it found the result in uh 64 seconds it was a perfect match and it um it actually invented a relationship which is called thread one um which we can that's that's imagine that it's actually means connects to and it found a static Rule and a causal rule uh it said that a light is on at any moment in time a light is on if a light that connects to it is off and it found a causal rule that said um a light turns off if it uh connects to another light that was also off so that's how the the lights change over time the status of on and off and it came up with initial conditions that said that well a connects to first of all that there's an object one a light called object one that we don't see but is there we imagine is there that a connects to it uh the light object one connects to B and the light B connects to object one so that's the causal rule that it discovered now if we uh run uh this causal rule produce a Trace and as you can see the trace matches the observation it adds an a new object so the coverage being uh excellent the perfect in this case and our causal theory is a good one it's actually a perfect one it's no it's not necessarily the only one though so is this causal Theory Unified so going back to uh Dr Khan's requirement of synthetic Unity of that perception not every causal Theory will do though it may predict correctly it may not be meaningful unless it is Unified well is it we've saw the fourth dimension there's four dimensions of unification is it specially spatially unified well all our objects are connected directly or indirectly to each other that's good so we have spatial unification do we have conceptual unification so we have this new predicate we have two predicates right uh Fred one which we uh translate to connects to and then the predicate that says uh whether the light is on or off well we have a constraint that says that a a light can only be connected to one other light so Fred one has a constraint on it that says it uh uh it's

it's exclusive so an object cannot have read one to two objects you cannot connect to two objects that's a constraint of us discovered and part of the causal Theory and also uh implicitly um a uh the uh on uh relation of the predicate as the value on or off and it cannot be both at the same time so it's conceptually Unified is it statically unified or are the rules of static rules obeyed well uh for example the static rule would say that uh given that b uh connects uh to a if B is off then a must be on so if you look at any uh place where B is off a is going to be on and you could do that for uh every uh other uh light and relationships between lights so they they all obey the static Rule and the causal rule says in for example here that if B connects to a then if a was off then B must turn off so if you look at uh B let's say e or B was off yeah if I'm sorry if uh B connects to a and yes and if a was off B must be off so if a was off b b becomes off the next step so that's correct as well so statically uh we are true and temporarily we are true we are unified and of course that uh we get a thumbs up from uh Dr Kant our causal theory is Unified thus makes it makes sense of the observations of the sensory inputs now it's no accident uh that um uh can't would be with figure in an active inference project there is a link uh between active inference and Kant and it runs through the the celebrated 19th century uh German engineer Herman Von ham Holtz he was a disciple of Kant and he developed the theory of uh visual perception that operationalized Khan's epistemology and in fact it anticipates predictive processing uh in 1995 uh Peter dahin and Jeff Hinton uh developed the helmholz machine name is in his Hunter um and it is a type of artificial neural network that's trained to create a generic model uh from an original set of data and it can account for the hidden structure of the data so there's a as you see there's a there's a link which is uh discussed and elaborated in this paper which is very very interesting paper all right so close parentheses so we've looked at the perception engine uh from the perspective of philosophy so uh now let's look at it through the lens of machine learning we the observations that constitute a training set it's a very small one and the opposition engine is uh the learning algorithm and what is learned the output is a causal Theory um so the the learning process is uh is unsupervised logical inferencing and the um the output is a human readable logic program so we see here that there's there's some profound differences with uh the more popular form of machine learning in that the training set is really small um that uh the uh the product of uh the the learning the uh is actually a human readable artifact in this case a logic program so this is the training set um uh as as inputted lights led a turned off at time one B turned on at time one a term on at time two be turned off at time two etc etc so that's the training set and you feed this into the um uh Obsession perception engines algorithm and out comes a uh causal uh Theory so in in a

little bit more details uh what the algorithm is and does first it extracts um the uh observed object extend objects the object types and predicates from the observations so we have on we have object a object B we have led as an object type so that's all part of the observations and that becomes part of the extent vocabulary then the opposition engine imagines uh unobserved objects types and and predicates uh for the relationships and properties and that becomes a latent vocabulary so there's a a step of imagination then using the combined vocabulary you know the both the extent and and latent vocabulary combined it looks for a unified causal Theory a set of constraints rules and initial conditions that uh obey the um the rules of um the constraints of uh synthetic uh Unity of our perception um once it has this causal Theory and with initial conditions it applies the causal Theory to these conditions and produces a Trace uh it recreates observations if you want and and augments them and and extends them into the future uh now it looks at this trace and Compares it with the initial observations for coverage and decides if this is a good causal Theory or not then it also looks at the causal Theory complexity how many rules how complex are the rules Etc and um measures for complexity so if we have a choice between two uh causal theories of equivalent coverage um the app perception engine will uh select the least complex one using Occam's razor now if you look at this algorithm you'll see that the the the boxes in green are um not uh deterministic that's this is where search happens we can posit different kinds of objects we can find different kinds of rules so this this was this is where search happens now app reception is implemented using logic inference actually it uses uh three forms of logic inference there's the one that we're more familiar with which is uh deduction where given rules and causes we we infer the effects then there's uh induction we're given causes and effect we look for the rules this is what science does right looking for rules that would account for uh effects given the causes then there's abduction where given the rules and given what we observe the effects we're looking for the causes in this case we're looking for the latent objects the latent or relationships between these objects and then you can combine both abduction and induction where you're given effects essentially observations and you're looking for both causes and rules which is what the app reception engine does and this is where in the um in the algorithm uh these kinds of inferences uh are at Play so positing latent objects that's that's a form of abduction uh imagining causes finding the rules well that's clearly a a form of induction and then applying the rules to uh uh of a causal Theory to um some initial conditions to create a trace well that's that's deduction we have the causes the initial conditions we have the rules causal Theory and then we produce a trace the effects so that's that's deduction so the app reception engine uses all forms of

logical inference and and now just a reminder that um the output of the perception engine that is what is learned is actually human readable uh you may want to compare that to a large array of floating points uh produced by a traditional the more uh popular form of machine learning nowadays so here this is what's actually produced by the app reception engine as it runs on a set of observations it produces um a logic program that is human readable the only when you look at it the only thing you need to to kind of guess is what does the what what is meant by pred one and if you think well maybe it means connects to maybe the lights are connected underneath uh a board out of sight of um of the Observer but finding a unified causal theory is hard so we have to guess what the latent objects and predicates are what what are the hidden uh lights what are the hidden relationships between lights and we have to discover what constraints might apply on the predicates um and what are the initial conditions from which we want to recreate a trace uh what are the static rules that apply to a simultaneous observations and then what are the causal rules that given uh observations at time T will predict observations at time t plus one this this is hard as a matter of fact it's a non-polar polynomially hard uh the search space grows exponentially with the size of the input which is the the the the size of the uh accident extent and latent vocabularies and so just like in chess you can't uh predict to the end the consequence of a move because of Contour explosion with the app perception engine you cannot systematically Traverse the entire space of possible causal theories to find a good one because it's impossibly large so the job of the app reception engine is to find a causal theory in a ridiculously large haystack how to do this in my implementation I follow the uh the the the recommendations in in uh and and I follow also the implementation in Richard Evans paper by breaking the search space into uh chunks first I there's a region and the region says um so how many latent object types objects and predicates will we allow so what is the limit of imagination of the cognition actors that is trying to uh perceive a causal Theory what are the limits of its imagination and within that region um of bounded imagination um we carve it into templates where we say okay we're going to use these latent objects types these latent objects and uh so basically what vocabulary specific vocabulary we're going to be using we're going to use object one we're going to use object uh thread one on top of the uh observed on uh predicate and observed A and B lights and we're going to set the maximum complexity on the rules and see if within uh we can find causal theories that fit this template so this is a carving up of the search space and having broken the search space into regions and templates uh we have Scopes in which to apply heuristics now why heuristics because the systematic traversal in it cannot be done in reasonable time there's

just too many uh candidate causal theories to look at to find a good one so we use heuristics we find we find ways of maybe getting to a good solution faster at the risk of missing it but at least uh we'll have an answer or no answer in a reasonable amount of time and there's a number of of heuristics that I have implemented in my implementation of the perceptron engine well there's time boxing at some point you'd spend no more than this amount of time looking into a region or into a template uh there's multitasking well the problem is actually as they say embarrassingly parallel you can explore multiple regions and multiple templates in parallel and so make good use of a multi-core uh computer uh you want to make sure you don't repeat yourself so you don't want to uh Traverse the same region uh twice or look at the same puzzle Theory twice you want to satisfy you may be a good enough theory is just fine we don't want to look for the perfect one necessarily we may not have time um you want to fail early if you're in a region where nothing good is found you may want to leave it uh quite quickly at the risk of maybe not finding the a good one that is just over the horizon but you want to be impatient you want to throw the dice you may want to kind of mix it up so that every time you run the Apple session engine on the same problem uh you may find a solution a a different solution first um you want to uh go for The Simple Solution first you may not want to try everything to sample sum we want to start with the the easiest uh part of the the search base first be judicious and so forth and so on and most importantly be selective so reject any uh causal theory that would fail the the constraints of unity of our perception with all these in place on the running my implementation of the app reception engine gives pretty good results so here I I did a run this is not Cherry Picked I said decided to do one a series of seven runs uh and collect the data and show it and in this run I set up the app position engine to only accept a perfect causal Theory one that would uh produce a trace that totally covers the observation so and I did seven runs uh the first one succeeded found it in four seconds uh the second one it took 102 seconds so there's there's some randomization in in the order in which things have searched is luck is involved as I said um the third one one second that was that was pretty cool the the the fourth one well took 204 seconds uh then 96 12.99 so uh what a a good distribution here now I said okay I'm gonna I'm gonna run the perception engine again on the same uh training set that I showed earlier those two lights but this time I said um I'm gonna be uh I'm going to accept the theory that has at least that has at least 80 that has 85 or more coverage so it it predicts it it recreates the the observations well enough but not perfectly and I uh time boxed it to 30 seconds so you have 30 seconds to find It Go um the first run it signed a a causal Theory with 75 percent uh accuracy uh immediately then the

same accuracy same coverage 10 seconds 11 seconds it hit 80 29 seconds it found a perfect one then 11 it found 87 coverage and stopped right there that's good enough 75 again 100 is the first one it found above 85 in 18 seconds and 75 zero seconds so a good distribution again so we're getting into uh reasonable times we're not talking about hours here we're talking about seconds and I'm I'm hoping to um do further optimizations and bring it down to something even smaller so that a cognition actor can say uh uh I want to make sense of these observations query the app reception engine and get an answer a causal Theory within uh maybe a couple of seconds That's My Hope now something interesting sir it so happens that what makes it hard for the app perception engine to find a good causal theory is formally equivalent to what makes cognitive science as a whole hard and that's this paper here makes the case uh and proves the case uh quite quotiently um so um the cognitive science wants to find models functions or algorithms that explain account for um situated behaviors so you you feed in uh you feed into this this the cognitive science machine um pairs of situations and behaviors and you want to come out of it a a model an explanation a a function or an algorithm that accounts for it well the the paper makes the case that if if if the explanation is is to be bounded in size then the the the the problem is computable but it's not tractable in the same meaning that it's combinatorially explosive but once you have a solution it is uh computable and tractable to verify that the solution is good that it accounts for the um the uh the data that you're trying to understand well this is this is equivalent formerly equivalent to what the app reception engine is doing my implementation was done in prologue uh I will not go into the details it's about a thousand lines of prologue I'll just say that prologue is a programming language uh that use deductive inference as its model of computation with backtracking um so essentially it searches it searches for a solution and will backtrack if it took the wrong um Branch if you want it and and and we'll look for a different way of satisfying a a line of the program um so let's just say that it it makes traversing a search space uh it gives we get traversing a search points for free when we program in prologue I won't go into any more details but you can see some prologue code here and it's uh the fun thing is that um a a prologue program is akin to a logical description of the problems trying to solve I think it's very cool and the prologue environment was augmented by something called constraint handling rules which is an extension to prologue that adds abductive reasoning so basically in the program you can say assume this is true until proven otherwise and the um the uh the rules the CHR rules um are there to uh verify if it can be proven otherwise so um again I'm not asking you to understand this quote at all but I want you to realize that this code is the code

that actually executes a causal Theory um both the static and causal rules to build traces it is that small it's it's very powerful so combining prologue in CHR um I found extraordinarily uh powerful and very excited about it I'm a programmer next steps well next steps now that we've solved uh individual learning by cognitive uh well I want to move to uh beliefs from from the sensations and to policies to validate or eliminate beliefs um a lot of these beliefs actually fall out of our perception latent objects and latent relationships and properties are can be considered by as beliefs then there are other kinds of beliefs that can be uh obtained from uh what's been not perceived um they'll be introspective beliefs uh that uh uh communicate um how the cognitive cognition actor is doing in terms of competence predictionary rates um how well uh it's um its our perception is doing is and whether it is uh engaging with other cognition actors is it relevant I will have feelings which will provide normativity to these beliefs so if uh feelings are signals of uh risk to homeostasis uh loss of resources physical damage uh too many prediction errors so that's hunger pain and fear uh and feelings will take beliefs over time tainted beliefs and Norm good beliefs bad beliefs will want to be eliminated or validated through policies um that will be synthesized by the cognition actor and each cognition actor Will emit uh make available to others um its API what what predictions can be made about the beliefs of this cognition actor what actions uh are available to others to be uh uh asked of the connection connection actor and then as cognition actors uh uh connect to one another uh as the uh Junction actors formed the um belt of other cognition actors then uh the a construction actor will be able to predict the beliefs of others will be able to uh compose policies made out of actions that are implemented by other cognition actors and eventually um we'll have a Society of Mind which is a bunch of intersecting uh boom belts so um that's uh it um so I see what uh the Society of mine is a complex system uh of collective theorizers and I'm gonna try going further with this project to answer the question if uh Collective of theorizers can self-organize to actively sustain itself so thank you uh to the active reference Institute for inviting me to present and for providing a home for this project and for the constant support and encouragement I'll see you later on Discord thank you awesome thank you JF just to conclude the session I'll I'll read two questions and let's maybe address them in an upcoming Robotics and embodied meeting so if you're excited about this project certainly we all are and about symbolic active inference join the Discord and participate in the Robotics and embody it but I'll drop these two questions from David Williams in the chat who wrote One how important is conducting this work in Real World versus simulation and two what tools or components are missing in the robotics toolkit to make this research easier and better I know those are things

you have a lot of thoughts on so I'll look forward to uh discussing with you more thank you JF
thank you peace all right see you and it's a great Segway from Collective behavior in Surprise
minimizing agents two Collective behavior in Surprise minimizing agents I guess welcome
Connor hey sorry I just turned on my audio so I just heard the last thing you said yeah okay
both can hear you yes thanks we're looking forward to your presentation multi-agents active
inference and multi-scale Alignment current developments and challenges so feel free to share
your screen or proceed however you prefer great thanks and before I start can I just um
because I've had issues with my my voice like saturating like my microphone saturating how's
my audio is it clipping at all or does this sound okay this this is good and I'm watching okay
perfect thank you um I will share my screen I'm going to share this one and then the one thing
I'll note oh yes okay it's just JF symbolic implementation does not use statistical distributions
it uses the um symbolic and The Logical inference and now we're gonna move back into the
distributional space and it'll be awesome to see similarities and differences so thank you
Connor to you for the presentation and thank you um Daniel for the introduction and for
inviting me as well as big thank you to the other organizers of the third applied active
reference Symposium um yeah I'm really happy to be here to present so my name is Connor
Hines I'm a PhD student at Don much Park Institute of animal behavior and I'm also a
researcher at the versus AI research lab rmd Lab um so I'm gonna do something a little bit I
guess unconventional for people in my position like I'm a junior researcher coming to the end
of my PhD so usually when I give a talk I would present on my own research um like what
I've been up to for the last 10 years working on but instead of that I'm actually going to talk
about given the motivation of the Symposium I'm going to talk about something that's more of
an overview or perspective on the current state of the field in multi-scale active inference or
multi-agent active inference and what in my opinion we need to do to move forward as a field
I think that's very um resonant with the kind of motivations and the title indeed of this
Symposium so I'm going to give a general analysis of what multi-scale activations is why it's
import important I'm going to provide a brief analysis of its formal basis as it currently stands
and then what we need to develop in this kind of sub-discipline of active inference multi-scale
active inference to really make it rigorous and really to actually reap the benefits of what it
promises um so generally active inference has been used a lot to design agents that can solve
problems plan and just generally emulate behavior that we deem intelligent which includes
things like Risk sensory decision making intrinsic motivations to resolve uncertainty and
finally from a more scientific standpoint the ability to firm a process theory about how

biological brains actually might work um but in a lot of the theoretical work on active inference from the last 10 years is or 10 plus years really there's also alongside all the kind of practical building adaptive agents there's a claim that active inference is inherently or intrinsically multi-scale from the very get-go it is a multi-scale framework it's not just about building single agents um so it's really whenever we write down a single active inference agent what we're implicitly implying is also a nested hierarchy of active inference agents both below and above so colloquially you'll often see this in papers as the idea that there's Markov blankets all the way down markov blanket is a statistical structure that's very kind of um intrinsic the definition of Agents as they are defined in under active inference so I'm not going to get into defining that I'm kind of assuming that there's a more disciplinary audience there but I'm sure other touch feelings Symposium can provide better Clarity so yeah Mark Hawk blankets active in difference all the way down all the way up and at any given scale the free energy minimizing Dynamics or the active inference Dynamics are kind of claimed to be aligned with or parallel to the free energy minimizing gradients at the level below and above so what what the claim is that as agents are doing their thing and doing active Infamous at one level it both entails and is constrained by active inference processes of the macro agent that they're participating in so I'm a cell that's part of a tissue as well as the microagents that comprise them I am a free energy minimizing cellular agent comprised of organelles that are also minimizing free energy so this this kind of constrained neat um nested gradient descent on free energy is part of the story of multi-scale active inference and it also crucially assumes that these Dynamics are aligned correlated Cooperative across these different scales so I should mention that there is a formal argument made more recently I would say in the last five years about how this is possible and it relies on an apparatus from statistical physics called the regionalization group this basically allows you to analytically identify shared symmetries energy and conservation laws at different scales in a given system that's comprised of subsystems and subsystems so on at infinity so there's a formal argument specifically made in the a free energy principle for a particular physics monograph from 2019 that applies the renormalization group apparatus to multivariate stochastic differential equations that are kind of the equivalent point of Ages so you can apply that that framework to certain sorts of coupled stochastic differential equations that exhibit Markov blanketage Swatch coupling structure and you can kind of prove analytically that there are going to be nested systems of Markov blankets and that they're all of some sense of minimizing three energies at their own scales so I'll get more into that argument later but I just want to mention that as I Define multi-

scale active inference that um that there is a formal argument that that's related to it so this slide I just put together to demonstrate the idea of nested free energy minimizing processes visually so at a given scale we can think of an agent as occupying some point in its free energy landscape indicated by this Red Orb um which represents say its configuration its beliefs and its actions and it performs active inference and in doing so minimizes its free energy so it changes the position of that ball on that landscape and that is all we mean when we say active inference that that corresponds to the agent doing inferencing doing action and kind of getting to the fixed point of its local free energy landscape the multi-agent case is simply when we add more of these processes so there's other agents usually assumed to be similar agents and the word similarity let's put the asterisks on that and they're all sitting at different points in their own free energy Landscapes they're all the the position of their local Red Ball is maybe in a different place so the claim of multi-scale active inference is that as we link these multiple active inference agents together so they can actually exchange information like observations and actions with each other what we automatically get is some kind of super agent that is also minimizing variation of free energy and is in some sense an emergent or supervenient active inference agent and I say the word we automatically get a super agent with an asterisk because there may be some conditions on that mapping from local to Global that have to be elucidated um so we'll come back to that in a bit but in short I think the the definition of multi-scale active inference is very eloquently put in this paper by um Raphael uh Kaufman pranav Gupta and Jacob Taylor I think Rafael Kaufman is actually going to be on the panel later um and this line from their paper is a really nice uh I think just summary of what it is so I'll just redo that right the upshot is that in theory any active inference agent at one spatio temporal scale could be simultaneously composed of nested active inference agents at the scale below and constituent of a larger active infatation at the scale above it in effect active inference allows you to pick a composite system or agent a that you want to understand and it will be generally true that both that that agent a is an approximate Global minimizer of free energy at the scale at which that agent reliably persists and that agent a is composed of subsystems a sub I that are approximate local minimizers with free energy so that is the claim as I'm going to like continue continue evaluating in this talk and I think it's just a great reference point to make so okay that that that's what multi-scale active inference is why is it important why do we actually care about that that sounds philosophically nice and beautiful visually but why is that important so there's a ton of actually really important implications of this both for the engineering and the Natural Sciences um first of all the namesake of the Symposium I assume

inspired by this recent paper by Carl ferson at all about enacting ecosystems of you know parenthetically shared intelligence so this is the third applied active Symposium so to to really make this resonate with the applied aspect let's make this very concrete if we can figure out this multi-scale endeavor then we can actually engineer distributed systems of multi-aging intelligence where local agents and doing their own little local active inference processes are also cooperatively instantiating a global agent that's also performing active interest this has huge computational potential of course compared to kind of the state of VR predominant methods for artificial intelligence which are deep learning which really is about propagating Global Information through an entire computation graph so although you could argue back propagation is local in some sense it's really not local in the way that multi-scale active inference promises to be local so if we can figure out how to actually engineer multi-scale active inference it will have really tremendous implications for the study of artificial intelligence just from that pure engineering standpoint it'll be cheaper in one where it will be computationally energetically Memory wise cheaper a lot cheaper um secondly from a kind of more Natural Sciences motivation which is kind of where I'm coming from I get a piece I'm doing a PhD in biology so I'm interested in questions about actual real systems in nature just the idea of being able to get super specific and rigorous about phrases like emergence intelligence emergent computation collective intelligence super organism that's often thrown around we're talking about social insects right these are terms that you hear thrown around in many different scientific disciplines that deal with multi-agent systems um Network systems but none of these terms to my knowledge have really rigorous or precise conditions multi-scale active inference is a kind of framework that's in the position to provide those rigorous definitions and conditions so from a scientific standpoint it could really lend a lot of potential and usefulness for other scientific disciplines and finally another pragmatic motivation there's loads of fields that are obsessed with designing and Engineering Systems where local selfish individual Behavior can lead when networks appropriately to some desired Collective outcome and these disciplines you really want to figure out how to engineer that properly so this goes from the design of financial markets and trading systems all the way down to how do you design a multiplayer video game so that that's kind of just motivating why is multi-scale active inference even interesting so then the question of course becomes is the multi-scale active influence clean actually true are all multi-agent active influence systems comprised of and themselves comprise nested hierarchy of free energy minimizing agents a glance at a smattering of other scientific disciplines that specifically deal with multiple agents multiple

interests Collective phenomena like coordination group Behavior collective intelligence I'll go answer all those disciplines would naively suggest that the answer is no so there's things like frustration in thermodynamic systems Game Theory the very existence of zero-sum games and Nash equilibrium bandwagon effects when we're talking about social networks and and um opinion Dynamics um sacrifices for the common good which we see in different contexts in biology like in the context of kin selection but also in the context of arguably cell cell death in a tissue um these are all basically plenty of systems where local constraints and Global constraints or desires or free energy gradients whatever you want to call them coming to direct conflict so the obvious example that I have listed at the top of these on bullets is the idea of geometrics frustration that we see in ising systems with very low temperature so these izing models basically describe lattices of ferromagnets that are happy when they're pointing in the same direction as their neighboring Fair amendments and they can be in an up or down state so basically the magnet can be pointing up or pointing down and these Global Systems are defined by a global energy function and the whole system is in some sense trying to minimize that Global energy function but sometimes you'll find cases in these Collective systems where this little spangle in the middle cannot be happy because they're getting conflicting information from two neighbors I want to be pointing up in blue like the agent on the left but I also want to be pointing down like the Asia on the right so this is a system that's collectively finding some fixed point of its Global free energy but it's actually leading to a local conflict where this agent is not at a point where it can do anything to make itself happy or minimize its free energy further so just from even the basic glance at a fair magnetic lattices we already see instances where local and Global gradients or local and Global Optima are not aligned in the right way so given all this the burden of proof for multi-scale active infantry skill on us so we need to show that Collective active income systems generically do align again across scales and maybe if we can put an X across the word generically and it's not some automatic condition if they don't then at least we have to establish exactly the conditions in which they do so anecdotally we do actually have some conditions there's some seem to be some kind of basic ingredients to get Collective Collective active inference to work so one is that we basically need agents to exchange actions and Sensations across some kind of Markov blanket this is not really a conditioning this is almost more part of what it means to be an agent so having more comfortable and get separation between agents is just another way of saying we have multiple agents in our system rather than a single agent if you're violated a Markov blanket property so internal states of One agent are not allowed to see the internal States or

external states of another agent then you're kind of cheating because you're kind of saying it's actually One agent and what you're doing is information sharing within the brain of the similar region the second condition which is something that's often hallucinated more anecdotally and not really formally is this idea that agents need to have some kind of shared narrative or shared hidden States or sensor space in their generative model so I've worked a lot on Collective active inference systems just simulating agents and trying to get them to do interesting things together and my intuitions and experience do agree with this basic fact if the agents don't have any similarity in what they're representing or trying to achieve then it's kind of like trying to fit a square peg into a circular hole so this is really nicely elucidated in the one the earliest cases in this paper a duet for one by first and first from 2015 where they show that for two agents to really align they kind of have to have a shared generative model and then you can get kind of this nice synchronized Behavior again though these things like what does similarity mean what does a shared narrative actually made formally mathematically those things have not been initiated yet so right now a lot of the building of these Collective systems is based on our intuitions and kind of engineering things in using some vague guidelines like oh they should have a shared sensor space but there's no mathematical conditions or guarantees about what degree of similarity is needed between two agents models to get the intended Dynamics and finally we have to have at least some agreement between the generative model of each agent and the generative process which is really the behavior of the other agents generating their data so this is kind of related to the the previous point about having shared generated models but just to be very specific the physics of the space that transfers your actions to my observations that physics can't be dramatically crazily different than how our generative models represent those physics so if we took two fish with the same generative model of each other and they normally would school together in a fish tank but we throw them in a volcano or shoot them out into outer space they won't school together because then the generative process is so dramatically deviating from the way they're representing that physics the the way they're generated model is constructed so these are again just ingredients kind of guidelines or anecdotes but there's nothing really rigorous Behind These conditions they're more like a list of best practices so now let's get on to actual rigorous stuff so the first real rigorous attempt to show that multi-scale active inference generally works is in one section of this free energy principle for a particular physics monograph from 2019 so it leverages this apparatus I mentioned earlier the renormalization group operator to basically show how one can successively course screen multi-variate stochastic differential equations that admit sparse

coupling between their state variables so the main result in my mind that connects these renormalization group results to multi-scale acts of inference is the fact that the lagrangian of the system at one scale can be expressed as a function of the lagrangian at other scales and that applies in a scale invariant fashion that is the main like uh kind of output or the main benefit of using a renormalization group apparatus so you can kind of think of LaGrange like the generative model it's a physics term but it's related to the generative model of the agents that comprise the system and therefore also their free energy so in terms of active inference it means that this reasoning of the renormalization group can be used to smoothly move between the models of individual agents at one scale and the model of a collective or larger agent at a different scale or a smaller scale for that matter and the nice thing about it is general for all kinds of Dynamics and thus generic models it doesn't depend heavily on the form of the stochastic differential equations that form your um the issues with it is that there's not a Global Link to basic mechanics and active inference it's still all done in the traditional physics formalism so we don't actually have an explicit link to local inference and Global inference although if you know the connection between the lagrangian and the generative model and the free energy then you can make that connection but it's not actually made explicitly for us in this part of the monograph it also requires the assumption that the generative model and the generative process are identical at the local level that's where we need to have lagrangian is defined that's also a restrictive assumption that's probably not realistic in my opinion and then finally there's something about uh kind of spatio temporal segregation of scales so we need to make assumptions about how fast random fluctuations are at one scale relative to another scale in order to justify kind of core streaming or forgetting about certain States As you move between scales and that's also something that you could argue current Research into Collective Dynamics challenges that assumption about how how fast noise is at one scale relatively to the max so now I'm going to discuss quickly another small contingent of active inference research that is attempted to address this mapping between local and Global Inference processes so what I want to kind of just generally say with this presentation and to our community is that the kind of approach taken in these two papers first of all active inference model of collective intelligence and speaker glass systems there's Collective active influence this is one of the types of research I think we really need to move active inference multi-scale active inference forward so I'm not trying to be too biased because I am the first author on the second of those papers but I'll also be the first one to point out the limitation but benefits wise I think these approaches are really important because they formally relate a local generative model at one

scale to a global generative model at a different scale so really Ty how did the parameters of one model relate to a coarse grained model and these are really good um steps in the direction of a formal theory of collective intelligence that goes from local intelligences to Global however there's still issues with these one of them is that they only deal with the issue with inference at the global level not active inference so these both these papers concerned with a bunch of local active inference agents that cooperate to form a global inference agent like a passive-basing agent rather than an active inflation and it's also unclear whether the system studying newspapers are actually very generic like the results are generic to study collective intelligence in general or their nice formal arguments but they're only applicable to these specific systems so we still don't have something that's even more kind of zoomed out and Abstract than these which tend to be a little bit of case specific and finally the actual um scale Transcendence that we're doing in these papers is still relegated to really one step we're not doing the full multi-scale infinite scale regression that's something like renormalization group so that's kind of the current overview of what I think are the most promising directions in multi-scale active inference and I'm aware on time sorry five minutes so I'm going to quickly try to go through what I think are really promising directions to push in terms of multi-scale active inference and kind of intuition products that I think will help us study these systems in a way that's different and also actually better informed by other research disciplines so the the general idea that I'd like to put forward is that misaligned gradients can actually be a good thing so it's actually sometimes good when local free energy gradients are misaligned with global grains so sometimes the global system will actually do better if the local systems are performing worse so this is something you could call multi-scale conflict with a free energy minimizing process at one scale are actually doing bad quote unquote but it's because they're being driven by some higher scale process that is doing well so rather than trying to always avoid constructing processes like this I think this kind of frustration to use the analogy from statistical physics can actually be an inspiration for what we should investigate further because it actually might be key to facilitating optimality at different scales um the reason I put this forward is because there's loads of research just Recent research in the last several years that are suggesting that actually making local agents more frustrated or more unhappy might coincidentally or not coincidentally lead to better Collective or Global outcomes so this is expressing various forms in various bodies of work but one of the biggest patterns I've noticed is the study of collective Behavioral Systems over the last several decades is the idea that local noise and local dysregulation can often facilitate Global coherence or the global coordination

and where multi-scale active inference has something to say in my opinion is in framing this benefit of local frustration in terms of a misalignment of free energy gradients so maybe that actually temporary misaligning local free energy energy gradients from Global ones may facilitate the dissent to fixed points in the global free energy landscape that satisfy everyone at all scales so I'm basically expressing an idea that's been known in various communities like stochastic optimization and stochastic resonance theory for decades but I think we as active influence practitioners have a new and potentially useful perspective to shed on that using the language of active inference and free energy and minimization and Bayesian inference in general so instead of thinking of accelerating optimization by just adding noise to the system we can think of exactly how to design local generative models such that there's an optimal misalignment of local and Global gradients or local and Global generative models in a way that facilitates everyone in the long run actually facilitating or minimizing your free energy um so yeah that's just kind of like a something I'm putting out there I'm investing in it now in my own work but I have no real results on that but I just want to put that out there in this venue because I think it maybe will inspire other people to think in a similar way so just to conclude now multi-scale active inference I would say is still largely based in theoretical or philosophical descriptions and illustrative simulations but we're still lacking a formal Theory um there's some theory in terms of the renormalization group arguments of the monograph but they're still in my opinion a bit underdeveloped a little under demonstrated and relying some restrictive restrictive assumptions like the fast and slow fluctuations the identity between generative monogenic there's a few more recent papers those two by um Kaufman at all and then by myself on spin glass um systems that have attempted particular proofs of multi-scale Bayesian infant systems in particular situations but their generality is still not known and not proven so we why I'm kind of trying to conclude with is by saying we need to incorporate findings from other disciplines related to the role of noise conflict and frustration in facilitating not subverting collective intelligence or Collective coordination and I think we can really benefit by looking at these other disciplines to help us build a really powerful formal creative multi-scale active inference and finally I think we need to set the goal post for what counts as a formal proof of multi-scale active inference and once we get there once we say okay this council's proof this is satisfying how could we use that to actually do the the hardest part in my opinion which is engineering actual multi-scale active inference systems that are intelligent and minimizing free energy at multiple scales um yeah so with that I'm going to conclude it looks like I'm just on time so yeah thank you again for the invitation to present and

I'd like to thank a bunch of people who are listed here and Beyond who have influenced my thinking and kind of my opinions um yeah if there's time to take any questions of course awesome great talk I'll just give a few seconds if anybody wants to type in a question also it's really cool like aswin in the previous session was highlighting pi mdp and just the way in which we enact the collective intelligence different people seeing a paper where an analytical formalization is introduced and then there's still so much work to get it to the package and then so much more work to take it to the Last Mile and I think your presentation really checked a lot of those boxes um I'll just read a question and then um that will just be an appetizer for our continued discussion so Marco Lynn asked do you expect the inferentially connected Dynamics to exhibit Behavior akin to theories of multi-body systems and to what extent can we transfer insights from that multi-body of work and then second question just for our thinking and learning from Marco have you explored integrating work with uh integrating work on self-organized criticality with multi-scale active inference or other Frameworks who can provide more flexible Frameworks or assumptions for a generic notion of multi-scale Dynamics great questions I hope that we can continue um have you back anytime or just continue to collaborate in the ecosystem so thank you for the Epic talk Connor and good luck finishing your PhD thanks a lot Daniel yeah I'll talk to you soon talk to you soon so thank you for that greetings all right well our next session hey hey Bert greetings great how are you doing good good very well um our next session is with Bert device Dimitri bagayev and Bart van Erp it's going to be called towards user-friendly design of synthetic active inference agents and I know a lot of people are super excited to see this really practical and Cutting Edge work so to you Bert and just let us know how we can support okay great uh well is my audio good yep sounds good okay then I'm gonna share my screen uh I hope I picked the right one I don't work it's Zoom quite often looks good it looks good yep all right super um well thanks a lot Daniel for for hosting this uh simple stream I've been watching some talks it's really amazing and uh we really we feel privileged to get the chance to present ourselves so we also just like a few others before is interested in developing a toolbox for active inference and so this picture or she kind of shows what we're um what we're about or what we're interested in so here's a lady on the left hand side and um I'm gonna try to get a laser pointer and she has this idea about a rewarding behavior for a vacuum cleaning robot right so she's writing down she has a textual expression move around the apartment apply suction until the floor is clean do not touch objects and when don't return to the dog so that that's not so hard I'm gonna rate that with one star out of three stars in terms of difficult difficulty level to specify that um but that's

not enough to program these robots right because what she really needs to do now is to specify a generative model and uh that you know the there's effectors and actuators right the robot has to move around to apply suction uh until the floor is clean so the sensors probably a camera uh do not touch objects so maybe there has to be object recognition this is a really difficult task to come up with this generative model here and on top of that she has to specify this kind of rewarding behavior in terms now of probability distributions of this generative model so very difficult I'm going to rate that with two stars because the next thing she has to do for this model is to specify the inference procedure to do actually active inference and free energy minimization in real time or this complex model and really that's almost impossible right only a few Specialists can really write a procedure for for variational free energy minimization in some very difficult model so what we are about what we've been working on is to try to automate the inference task so get rid of the three stars and yes she will still have to specify her model but in the long term we try to get away from that so in the long term we hope we will get a toolbox and now we're talking five ten years right um where a textual description would be enough to specify some initial model with an initial prior and everything else is just automated inference learning of States parameters structural adaptation of the model even maybe based on her feedback updating uh the the prior so that's long term for now we would be very happy if we could just automate the inference task um why why is it so difficult to specify inference for an active inference agents well we have so many competing kpis right we we uh we want to do this for large model Scopes not just for ABCD models but maybe there's also a continuous variables and our electrical models right it must be very user friendly you really don't want her to worry about robustness of her code um we don't want her to worry about whether um two variables are have conjugated relationships right um adaptivity we want to update States parameters maybe even the model the model structure has to be low power because these angels often run on edge devices all right so they run on their battery powered has to be in real time because you can't learn how to ride a bike if if there's no real-time reasoning and on top of that you actually want to minimize variation of reality right you want to do it at least as good or at least in a neighborhood of if people do a manual derivation and some of these the ciderata bite each other right if you uh you want to minimize variation free energy but you have to do it in real time and on low power yeah that kind of bites each other right so these are difficult kpis that we we want to yeah we are we they're all important you can't just take one out um because then the whole system wouldn't work um so um when you read papers on active inference you often also read and now we Implement variations of range

minimization and that can be done by message passing on a graph and I want to clarify first why it has to be done by message passing on the graph um I'll do that by giving a very short answer and then do an example the short answer is that Bayesian inference involves um Computing very large sum of products like what you see here on the left hand side here's a product $AC \cdot adbc$ and then we send them in AC plus 80 and so forth this is a sum of products now we know by the distributive law that this here on the left hand side can also be computed as on the right hand side If I multiply this out I get a times C plus a times D and so forth this is a product of sums and there exactly the same thing the only difference is that to compute the left hand side takes four additions sorry four multiplications and three additions to compute the right hand side takes two additions and one only one multiplication so on the right hand side is much cheaper to compute in the left hand side normally when we write down marginalization and Beijing inference we we write things down in the as in the left hand side what message passing does on a graph it will automatically convert that into much cheaper to evaluate product of sums and I'll give an example of that so here is an uh an example Model F of seven variables X_1 X_2 through X_7 and this model happens to be factorized F_A of X_1 fbx_2 and so forth now we can draw this factorization in a as a graph and what we do and this is called a Forney style Factor graph what we do is for each factor F_A we allocate a node so FB gets a note and FC gets a note and we associate the variables in our system with the edge and an edge is connected to a node if that variable is an argument of that function so FC is a function of F of X_1 X_2 X_3 and that means that FC connects to the edges X_1 X_2 X_3 and FD is only a function of X_4 so if D only connects to the edge X_4 so what you can see in this graph is this graph is nothing but a visualization of the factorization assumptions that we have for this model now if I'm interested in a big marginalization a task integrate out overall variables but X_3 so X_1 X_2 X_4 and so forth through X_7 I'm interested in this um then making taking advantage of this factorization I can rewrite this sum this basically this sum of product into a product of sums as as below here what you will see here below this computes exactly the same thing but I've made use of this distributive law for instance FC contains no X_4 no X_5 so I moved it over the summation sign to the left and FB also doesn't contain X_4 X_5 x_6 x_7 so I moved it all the way to the left and when you do that you are left here with an expression where I only some some over two two variables and here I have to sum over six variables and here over two and here over two so you can imagine if each variable would say so X_1 X_2 if each variable has 10 interesting values that you need to sum over then I have here the original marginalization problem I have 10 to the power 6 so a million terms and here in red I have 100

terms and here I have 100 and here I have 100 so here I have 300 terms and here I have 1 million terms so is it it's an enormous reduction in computational complexity when we make use of this distributive law now it turns out that if you write this out you can associate these these intermediate factors with messages on the graph it's just an interpretation a visual interpretation it's as if FC receives a message from fa and FB receive or FC receives a message from X from FB and computes an outgoing message μ_{X3} and the same thing for Fe so Fe receives a message from a neighboring factors FD and FF and computes an outgoing message or μ_{X3} so what you see here is that the entire marginalization process can be represented as basically Computing a few messages on a graph and multiplying some of these messages with each other and and and and that's how you can do Beijing inference and also how you can do variational free energy minimization so this works in um in factorized models but I would say even stronger if your model is not factorized and you have a lot of variables uh there is just no way you can do proper inference so any serious model is factorized like the brain is almost sparse is is almost empty we have well is it about 10 billion neurons and each neuron connects to a few thousand other neurons so if I would draw the graph that graph is almost empty it is hugely sparse and so there is no other way to do inference in the brain than by message passing um so that's why message passing just because it's more effective than anything else now then the issue is which message do you compute how do you compute messages um because there are different ways of doing it right and we also read in active inference papers you can do this by variational message passing or expectation maximization or belief propagation and variational LaPlace and all these terms it turns out that there is a an umbrella framework for all these message passing Frameworks and that umbrella framework is called constraint better free energy minimization and I try I will try to illustrate that by uh well by this slide so here I have this graph this is just an example graph where my generative model is basically factorized in $f_a f_b f_c f_d f_e$ and I've also written that here so this now is the variation of free energy now I haven't made any assumption on Q of X so Q of X is still Q of $X_1 X_2 X_3$ is just a joint overall variables and it doesn't have any factorization assumption it makes sense to also assume that the posterior kind of follows the factorization Assumption of the prior namely of the generative model so if we make that assumption and that means we're going to make the assumption that Q_X is also now a product of $q_a q_b q_c q_d q_e$ where $q_a q_b q_c q_d q_e$ stands for beliefs over nodes what I mean by that is that Q of B is a is a posterior belief over this node meaning it's a posterior belief over the edges that connect to this nodes just like F_B is a function of $X_1 X_2 X_4$ that's if you will the prior or the generative model then Q of B

the variational posterior for this node will also depend on X_1 X_2 X_4 and on no other factors if you just do that then you will count some of the variables double because X_1 is part of the belief over F_A but also part of the belief over F_B so we just have to Discount that by dividing by beliefs over edges that means that I make now an assumption that my posterior believes is is divided into local beliefs over nodes and local beliefs over edges over variables this will make things a lot simpler in fact if my graph is a tree and it is a tree here and I would do message passing on that tree and I could suppose I could do that perfectly everything is linear gaussian then I get perfect Bayesian inference there is no approximation so this is a good assumption sometimes it's still very hard to compute a message because even the the the the single message that come out of these nodes the still integrals or summations and in particular the integrals may be a problem we may not have an analytical answer so what we sometimes do is add additional um assumptions we'll say well the posterior belief over F_D I can't compute it in general but but I'm gonna just assume now that it's a Gaussian that makes it easier or we can make an extra factorization and say the posterior belief over F_B which is really I believe over the joint X_1 X_2 X_4 is going to be broken into uh um independent belief over X_1 and in the belief over um X_2 and X_4 these additional assumptions if I impose them as well this is what I recall now if I if I if I all substituted here in Q of X I get what's called a constrained beta-free energy this is the same deal as in the Oppenheimer movie this is Hans Bethe where it's named after and we have a graph now that that is highly factorized and we have local beliefs over nodes and over um over edges and they're indicated with with red and we have additional constraints in green there could be gaussians or I mean field constraints or other constraints and now we will assume constraint that make it possible to compute all the messages and now we can just automate this by making different assumptions we can turn this into propagation or um hybrid forms thereof we can turn it into any relevant message passing algorithm that you've heard of so this is a very nice umbrella framework that basically encompasses everything and and there is a we've written a pretty large paper on this um in entropy Journal where you can read all the math on how this works so we've talked about why message passing namely because it's the most effective way of doing it inference and we've talked about which messages to compute namely we turn our variational free energy into something called a constrained better free energy and then we can compute messages the only thing that's left is well when do we pass these messages right what is the sequence of messages which one comes first um this is where we see a lot of papers right you have the right control flow what's called control flow you have to say okay here is my algorithm for active inference

first I specify a model then let's do inference for every time step collect a new observation update the state update the desired future right and so forth compute expected free energy select the policy Etc um this kind of program the problem with active inferences is that there is nested for Loops in here here's a for Loop and here's another for Loop and for each of these policies I'm going to have to go into uh the future so I'm going to have another time Loop so it is for Loops in four Loops in for Loops this will completely explode in terms of computational complexity so as a result uh some very clever people have written very clever algorithms of doing this much faster sophisticated inference branching time active influence dynamic programming etc I recent proposals for doing this very clever in the end all of these proposals come down to a particular just a message passing schedule um once we commit to message passing on the graph as our inference procedure it's the only thing that's going on and all of this or sophisticated inference and granting time active inference all it does is it schedules the messages it says first this message then this message then this message I don't mean that as a slide to these algorithms they're very clever and as we've seen in the presentation by Asvane Paul you get huge improvements if you go from regular inference to sophisticated inference but it's good to realize that these algorithms just specify in the graph which message comes after which message um so here's an example here's an example of a graph and a sequence here's a message one then message H2 and message three goes up and then we go from FC to f and here's message five and then we go to Fe and this could correspond the sequence to Dynamic time programming etc or sophisticated inference um there are a couple of problems with this approach which we could basically with with having the user to specify a clever algorithm first of all you have to be a specialist to do it right only these are very clever people so so that means that um if we let it leave it to um let's say to an engineer in a company it's it's well it's the higher probability is not going to get it right that's very unfortunate um but there is another issue and that is that in it in a sense it's a global variable um in the message passing schedule all nodes are visited because if a node would not be visited then we shouldn't have it in the graph and that means if one node crashes basically the message passing schedule is invalid I have to reset my system and if you fly a drone if it's deployed and it's out in the field and a node crashes a transistor burns out and I have to totally reset now my system I have to compute a new message passing schedule um then you're not doing inference and you know and your drone flies into the wall so this is not robust and it also for very this the same reason we may actually want to take out a node we may want to prune you know we might want to do structural adaptation and we can't do structural adaptation because we have to reset the system

recompute a message passing schedule so this procedural style where an engineer app purely specifies which message comes after this message has some disadvantages it's not very robust and it's uh if you want to do it very clever you have to be really a specialist so a better system is what we call a reactive and it's very related to what was in the first session uh called the the actor model Keith Dugger had a nice presentation on the actor model so what we will do is we will say we will not have a global message passing schedule the engineer will not space specify anything anymore the inference code that an engineer will have to write is just say react to any free energy minimization opportunity in other words there is no inference code it's completely automated and um we will replace This Global message passing schedule by local triggering inside the node so each node is now just an autonomous autonomous system that's interested in minimizing its free energy it can do so by sending out messages and when will it do so well it receives messages I mean it feels when it looks at these messages and it feels like oh there is an opportunity for me to minimize free energy buy or expect the frequency by sending out the message then will send out a message and each node will do so by itself asynchronously so you get parallel distributed processing or concurrent processing as Keys call it um in principle you could play this game on many computers at the same time and so you get tremendous advantages first of all you don't have to write difficult code second of all you can do multi-threading or you can you can run it on multiple computers at the same time and so do and there's also um robustness um uh advantages because even though it crashes then there's nothing that stops the system from just finding another path right if if this node crashes this path from here's message three this path Now doesn't work so I cannot send anything to Fe anymore from uh no from X well then I just sent a new message here why not it's like when water falls down a mountain and it zigzags its way down into the valley and you have where you put up an obstruction it just finds another path um not the preferred path this has to find well the the second the second best path because the first path has been obstructed right and that's good that's what's going to happen in this system as well right that's just how nature works it tries to find the best path the easiest path and if that's not available then we do the second best pass and that's also what you can do with reactive message passing so you can throw notes you can do structural adaptation um and it's far more robust and you can also get uh let's do a chance encounters with other drones right drones that get close can start communicating with each other and when they are far away they stop communicating with each other and this is no problem because she can basically change who changed notes can change on the fly who they communicate to and who they want to listen to and uh so that's

that's the way data works and also how it works when we do a reactive programming and react with message passing so in summary we're interested in automating inference in um active influence agents right because it's an operation that's basically only for experts and this active influence technology is not going to be successful unless we get more people uh let's democratize it and we get competent Engineers being able to develop good agents right you shouldn't have to be a top specialist in the world to go to develop an active inference agent now in order to automate inference you must do message passing and I've talked about that for efficiency I've also talked about which messages to pass not necessarily do you have to follow this framework but constrained better free energy framework is very convenient it's an umbrella framework that basically goes over all the interesting other uh messages passing computations when Nas is passing a reactivity is passing it's fully automated so you don't have to write any code anymore in principle it's con it's you can do parallel distributed processing it's robust to structural changes you can learn new inference Pathways so lots of lots of advantages here how do we do it um I'd like to introduce a toolbox that we've been working on uh called RX infer and we do that with my lab here at the University I'm here in Eindhoven in the south of the Netherlands and we have a lab the lab is called bioslab here are postdocs and assisted professors and PhD students and we've been working on this for many years and some of these okay like Albert and Ismail and dice um have written dissertations um and our best work we have Consolidated that in a toolbox and the toolbox is called RX infer and you can if you want to have a look you can go to the website action4.ml and asking for our confer Works uh in the way that I've just discussed it does message passing it tries to minimize constrained better free energy that means it it can come up with all kinds of message passing algorithms um it will do it in a reactive way and it will try to do it in real time and low power and all the kpis that we're talking about now it's of course not done but um it's functional and I'd like to show some demos and I will leave it to Dimitri and Bart who are two Advanced PC students in my lab to show the demos so I'm gonna stop sharing awesome thank you great talk sure uh can you hear me yeah yeah okay I will try to share my screen okay so you should see it now looks good okay so yeah hello uh to everyone uh I'm uh Dimitri Vegas so I'm a PG student in bioslab and I'm from University of Technology uh and yes I have a small presentation about the actual software developments in bioslab and yeah so over the past few years we have significantly improved our tools and basically my entire PhD was dedicated to implement the this idea which bird was talking about uh like implementing the variational reactive message passing and in this presentation I just want to show you what you

can actually do uh using this Theory under the hood okay so basically in order to automate active inference right we need to automate Bayesian inference and we have uh we have already a lot of solutions for that as such as Stan pyro numpyro which is funded by Google internet is funded by Microsoft Turing is in Jewellery by MC and many many many so uh and basically these Solutions are really really good so uh and they're really good at prototyping as well but our goal is eventually to be able to deploy this kind of systems not just prototype and we are really focusing on this particular properties for this ultimate Vision inference uh so it must be low power adaptive real-time scalable it also must be user friendly at the end and it must support a large scope of models if you want it to be useful so and in biaslab we want to build such a software uh in such nice properties and it's always about Traders right so we do something better in one particular domain and maybe other software libraries they might be better in a different domain so but we are really focusing on this particular properties uh and so yes I will reiterate a little bit there's presentation so how do we achieve this so we have imagine we have an environment and we have an agent uh and the agent log takes some actions and the agent basically what he needs is to come up with some sort of good enough probabilistic model of its environment in order to do patient inference uh and in our framework we encode the model as a factor graph which not only models the observations but also actions and desired future uh and this this approach allows us to decompose the these complex relationships between variables and hidden States into some kind of structure and local blocks uh and it's not a block it's not a black box anymore um and the model itself may have some sort of background motivation interpretation it may encode your prior knowledge about some particular physical uh and the locality of these blocks basically allows you to scale to millions of variables and hidden States it allows you to pre-optimize it maybe or maybe use like some sort of different approximation strategies in different places so it allows a lot of very nice properties as well and we use reactive message passing to run uh actual variational version inference it uses reactive programming under the hood to minimize the approximation to the variation of energy and yes as berdos mentioned it's it's very much related to actor model and basically indirect transfer you can think of different nodes as actors themselves uh so and they have basically one single purpose is to send the variational message that minimizes the energy this is a very short and very high level description but it is essentially what is happening so we are not reaching different agents which interact with each other as actors but we also treat the the actual components of the underlying model as actors themselves it's like a very hierarchical structure so this is the main central idea of of this

inference uh so here for example first example we can do an inference in a dynamical system and this example which is quite old already I think it's like two years ago so we track a position of the object given some noisy measurements which are indicated by Green Dots uh the actual uh real signal we cannot observe it but we just can plot it uh it's shown as blue and the invert signal is shown as red and the data set is infinite the inference engine just reacts on it and does not assume any particular data size simply reacts in your observation as soon as possible uh yeah I'm actually not sure how smoothly Zoom shares my screen uh maybe you can see it's a bit lagging in the animations I'm not sure because maybe Zoom does not share it on a full frame rate um and also on the right hand side you can see how we Define models uh in our framework we use Jeweler as a programming language and so basically this is everything that you need to Define this particular model and run inference on a data set and actually I like really spend more days to plot it instead of in France right so inference was an easy easiest part from approaching was way much harder relate to user friendliness so and we actually have plans to improve our model specification language make it even easier so now for technical reasons we have some auxiliary statements and model specification language but we are working to improve that and make it even easier this is another example which is similar to the previous one that uses much more complex and linear dynamical system of the double pendulum and the system is chaotic and we can observe on this small part of it a lot of noise also indicated as a green dots uh and nonetheless given good enough model uh you can infer the other hidden states with pretty much High precision and the code needed for that is also relatively short uh and yeah we also have uh examples with active inference agents that interact with their environment uh so the left uh left up shows uh yeah Mountain curve problem very famous problem the left bottom side shows an active entrance agent which tries to control the inverted pendulum from falling in the windy conditions it reacts and wind we also have a demo of an agent that controls a pendulum in an ever-changing environment so on the right side you see a pendulum with an engine and Legend has engine has limited power and the agent itself needs to reach the goal and the goal is indicated as a red circle so and basically in this demo we can change the environment in real time and see how the agent reacts so you can change the mass of the pendulum on its laying for the amount of noise in the measurements or we can change the goal we can change maximum engine power Etc right so the agent will still try to infer the best possible course of actions in order to reach its goal and it just never stops reacting um it's it's also actually possible to restrict engine power such that it will not longer possible to reach the goal right

but the agent would still try uh we have other cool demos with smart navigation and collision avoidance uh which are still under active research and the code for them is not available publicly it will be soon available but for example in this example we can define a set of Agents with their boundaries and a set of their destinations so and we can see how they try to resolve their Routes all together um and we can have some static obstacles in the map we can see how agents can find their most optimal path in order to reach their goals and avoid any possible Collision and it's also not necessary to have static obstacles uh the obstacles themselves may move so in this demo we have hundreds of agents that navigate through a map of obstacles that move from bottom to top to the circles or obstacles and agents are depicted as small dots and they need to go from left to right basically avoid any sort of pollution uh and as I also mentioned we want to perform efficient and real-time inference but we also have to do it like low power low performance uh on low performance devices such as Raspberry Pi or coolpy as an example and we have some results of successful in running uh the Bayesian audio Source separation for example on cool Pi so it is actually possible we also run active we try to run active inference agents also on coolpy so as the aforementioned in word pendulum as I mentioned we also need to have a large model scope and basically RX infer has not been designed to solve any of the aforementioned problems specifically we have a large set of different examples in our repository uh different models different data different influence constraints uh we have examples for linear regressions hidden Markov Model Auto risk model hierarch models mixture models gaussian process and so on so right so this approach is very versatile and for example if you um if you compare it with sort of a conventional software libraries where you let's say have a library that solves a common filter might be a very great Library maybe super fast have top performance works great and very reliable uh super good but then you are constrained by by this particular model common filtering right and you can't really change it much we we are free to Define our own models uh which we can pretty much easily Define a model that essentially would act equivalently to Common filtering equation and so basically in the demo that I showed before about object tracking it was essentially a common filter that was written in a in a probabilistic uh so yeah that was my small addition to the best presentation so our software is free so MIT license and it's open source available on GitHub uh yeah and we would be happy thanks to be able to present where we would be happy to answer any questions thanks awesome awesome all right I'll just ask a quick question in the chat Marco asks sorry if I missed it are the collision avoidance demos real time adapting to other agents Behavior or is it a collectively pre-computed path um so basically they are not

uh super real time they're kind of fast to compute this path like maybe five seconds or so but it's very basically working to make it real time so we know what is the problem we know where to improve and we will make it fill that yes almost almost like next question do you have some comparative data with other methods and just more generally what kinds of benchmarks or what when you're talking with industry in different settings what are people like looking for that killer app of active inference or what are they looking for their key measures uh so I personally have a big paper about the comparison with sampling based methods like HMC and also in my PhD thesis cell there will be a comparison if not also other sampling based methods uh so long story short sampling based methods cannot really run uh this kind of sophisticated inference in real time uh they're very time consuming they do not really scale well to large problems which is really needed for active inference agents because if you have like a large environment very complicated you will have you will have a lot of unknown variables in your model uh so yeah so there is a paper that compares and basically we show that yeah our approach scales much much better so I personally run on just a just a regular MacBook laptop I run the model with 2 million uh unknown variables plus and it was like quite quite fast and then and if sampling based methods you may you may find yourself in the model with like 100 variables and then you wait like two hours or something and then it turns out that your chain did not converge so uh or something like cool yeah it's um people commenting in the chat like how far message passing and Factor graphs have come and so to to bias lab and to Bert at all we definitely appreciate this exciting line of research I mean there's so much to learn there and sometimes looking at the equations it can seem like it's like written in stone and just sort of the beginning and the end is you know variational free energy but then in your presentations you're really showing like no we we are Hands-On that's where we get the interpretability the modularity that's where it really is implemented and it's like an information Logistics challenge it's not like an esoteric philosophy question at that point no no in indeed I mean I I should say it's uh it's taken us I mean we're no Geniuses right so our lab exists more than eight years and and you see all the people in the lab it's taken us many many many years with lots of wrong directions to get this to work to where it's now so it's a very long path but the the at this moment I'm pretty confident that at some point in the future and I don't want to put a uh I don't want to say in three months or one year but we will be able to write a toolbox that will allow people to design a generative model and just press a button and and forget about the inference you don't have to worry about inference anymore it will be fast and automated and that will happen and it will happen within a few years and uh and maybe

somebody else will write an even better toolbox but I'm pretty confident that even our Toolbox will be able to do that so um and I think that yeah you know people talk about uh so why don't we have the success of deep learning and generative AI right well if they have the success because of a big data availability of Big Data big computers and toolboxes tensorflow and all the successes we don't need big data because agents collect their own data on or in the filter we don't need big computers active input agents you know they they manage their power resources but we need a really good toolbox because programming and active inference agent programming the influence by hand is just not doable so we need a really good toolbox that really automates this we hope uh RX inferior will be one of the first two boxes to do that I am sure that other people will also be working on it and and better toolboxes will come about um but I think the the optimistic message is that it will happen right uh and and once we have a toolbox like that then we can actually a large community can start building agents and we can actually show Deployable agents in the fields right that they work and they work better than the reinforcement learning agent or whatever is out there right um so that's I think that's the very positive and hopeful message it's what we expect it's what we prefer yeah yeah any last comments from either of you um comments from us no no we just I'm just very happy to uh well to get the opportunity and uh yeah I want to uh um everybody can download this toolbox um I think you at this moment you still should be a programmer to work with the toolbox and I hope you're friendly because you know it's not uh totally polished in the way that we want but it's coming right it's coming in the next years there will be a good toolbox for for almost everybody to use but people that are interested uh even people that are interested to work here at biceland we have open position uh for PhD students so we're happy to uh yeah to receive emails from people that are interested to have to work with us thank you Dimitri anything in closing no just thank you yeah again for possibility to present super nice to be here cool yeah well later in the year we will be discussing your two-part recent work and so we're going to be getting a lot into the details and I I hope that people in The Institute in the ecosystem will be as excited as we all are so thank you thank you bye-bye for the next interval we are in the section hosted by the active inference Institute on our recent paper active inference Institute and the active inference ecosystem so two short messages first if you're in the live chat I'll post the link right now to the paper and this can be more of like a conversational interval so if you have any questions about the um the ecosystem or The Institute or um any of the stuff that we're talking about let's make it a conversation uh and then second message if there are any co-authors on this paper who are listening to this stream then message me on Discord or post in

the YouTube live chat or just check your calendar and I expect some others to join in as we go welcome Pablo we'll be hearing in a little bit from you as a presenter is there anything that you wanna just bring up or reflect on before I share the screen with the paper let me just walk through it explore it and hear questions foreign I'll bring in the paper and again anyone watching live please just feel free to write any questions and uh I'm just going to bring it so that Pablo you're not going to see the paper uh but it's it's on the live stream cool well here we go this is a paper that had an awesome co-authorship and even broader circles beyond the listed co-authors of collaboration and contribution so just like everything that has happened and will happen in our work it was massively participatory and a real learning by doing and a real enactment moment the first version was released three days ago on Saturday August 19th and I'll briefly just list all the authors first we had the Institute as the first author so that the citation can always remain like Institute at all Institute comma active inference Institute and we also had Enrique John boyick Libra Burien Matthew Brown RJ corday Scott David David S Douglas Pablo Fernandez macquieta Daniel Friedman Holly Grimm Avalon kalu Raya Louisa Annika Virginia blue Knight Alexandra mcaleva Ali ramju Adil razi Jacob renan tamari Dean tickles and Alex vyatkin so this includes everyone from Founders and officers and board of directors and formal participants on different projects and different roles as volunteers or facilitators or interns and also people who are continuing to join the fun and show up and makes The Institute in our field what it is and what it can be what what we want it to be there's a table of contents for those who like to click through and just broadly there's going to be a preamble on active inference but outside of the Preamble this paper is not conceptual it is not technical it is not philosophical although perhaps it can be read in all of those ways but it's quite a logistical historical and vision oriented position document after getting some of the active inferences out of the way in the Preamble we're going to turn to The Institute itself in terms of our Vision Values principles and how we see moving through the challenges that we're facing right now onto the specific history of the Institute our current organization model in terms of our governance and Leadership structure as well as the Institute units the organizational units that support all the work we do we then turn out to the community and ecosystem growth and development and go through some details of different roles that are stakeholders and participants in our space and some of the tools that we use today for managing information and message passing we talk a little bit about our Communications plan internal and external both sides of the blanket and point towards some of the types of support infrastructure and administration that we do and that we will offer for

the active inference ecosystem we frame that in terms of continual deployment continual development and quality performance and growth evaluations across multiple scales close with some discussion and then provide an appendix that overviews a variety of the open source services that the Institute has provided over the past so let's just jump in I'm actually going to jump into the open source Services first and overview them before we jump back up to the top and look at the big picture because this is really to appreciate the work of so many contributors who stepped up in The Uncanny Valley in the gray Zone in the ecosystem as it's forming as it's always forming and really made it happen and so many people we know are out there scribbling or taking notes or printing out papers by friston at all or struggling and asking questions are there applications of this how would this matter why would this matter is there a code version of this how do you connect the dots between these equations like those are what we ask every day and so when people can combine their regimes of attention into a synthetic intelligence outcome then we go so far the first category of works that the Institute has developed are educational we've done video live streams since the end of 2020 till now more than 370 live streams all live streams are curated in terms of their transcript and we increasingly are developing automated and semi-automated methods for transcription translation sub captioning language modeling and so on we've had textbook groups actually now as of now we have three completed cohorts of the active inference textbook group The 2022 textbook we have two courses that are being offered that are produced by external content developers and both of these have only started in the last three four months first is Chris fields's physics is information processing and second is the work spearheaded by avell and co-organized with Kairos research active inference in the social sciences both of these are like awesome and great learning experiences as we heard from earlier in the Symposium we're working with Sanjeev namjoshi on his incredible textbook although almost to call it a textbook is to either think more broadly about textbooks than one has or constrain what it really is that he's doing because it's such a unique real research development full spectrum effort so I'm really looking forward to seeing how this plays out in 2024. until then for those who want to get involved with the testing that we do and we've developed the active inference ontology which is a critical resource for the ecosystem it enables us to Center accessibility rigor and applicability across languages and settings by knowing what it is that we're talking about when we talk about on the research side we've also developed the ontology in a research Direction in terms of research on and about and with active inference itself as far as software goes our primary developments first off has been the documentation and connecting the dots around

some of these other incredible packages that others are developing for example just in this Symposium we've heard from some of the core developers open source contributors to `Pi mdp` the python package for generative modeling as well as just a couple of minutes ago from bertofries and Dimitri bagayev coming from the Julia language `RX` and fur side and both of those are some of the leading packages for applying active inference so we do a lot of work to connect those dots and make it more accessible not everybody is going to jump in with all six feet to a and what we've done with active block prints with the work of Jacob smeckel and others is to wrap generative models from active inference inside of the complex systems modeling framework CatCat we've developed notation interoperability systems that actually let us pull back one layer from the implementational details themselves for example of `RX` infer or of `Pi mdp` so that we can bring it halfway to where the ontology and the visualization and the natural language exists and then use language models and other techniques to deploy it in an implementation fashion we've developed a host of research Publications we support people in all of their different stages of their career different seasons of life and some really creative work has been directly produced through an Institute project or somebody who knows may have gotten good ideas or good feedback just by hanging out being in the game and we've performed literature meta-analyses to help grapple with the scope and the quantity and the quality of what is unfolding before our eyes or before our antenna again even as we only so partially sample this ocean of information we're active in standard setting and continue to develop our qualification standards for active inference that could be the basis of Education research and professional development we provide a variety of mentorship and facilitation service type activities including volunteer program internship distributed facilitation and applied active inference symposia these activities just as all others that we're describing above are open source and they're also open to your participation so if you can follow some of those Suggestions by JF time boxing and prioritize and pay attention and be active all those fun things that he said and you're looking to do it in a semi-structured way with a volunteer program or in a more structured way with the internship then I heartily encourage people to get involved act now and in terms of governance activity we have a scientific Advisory Board as well as a board of directors and together with the officers those groups constitute the formal governance structure of the Institute but there's also so much more to it and we look forward to understanding what ecosystem governance can look like in the coming years so welcome Dean I just summarized the the work itself that was the pre pre Preamble and now if you'd like to say hi you are Pablo before we jump into more sections of paper or look at more questions

that people have asked how goes it or what was your experience of working with the paper or what brings you on to this strange stream at this opportune time uh basically why I'm here is because I for the first time even as as a acting board member with with the Institute I was able to get a better sense of the kind of depth that we aspire to and I'm not sure whether or not we are going to be I I really don't know what directions we're going to be going in the next coming months weeks years Etc but I have a better sense now as a result of participating in this development exercise of what the potential is that I probably had at any time and I've been with the active infant Institute for a couple of years so that's saying something I think we had to get to this point to be able to write this document and now it's it's acting as a bit of a springboard now which way we go and how we land I'm not still certain about what that looks like but I think that's part of what the really interesting thing is um we seem to have moved away from the dock and now the real Journey Begins so that's my take on it so far moving away from the dock d-o-c-k pushing off using it as a springboard continuing the adventure leaving the pre-print behind continuing to version it as a living document and also way finding outside of the dock Pablo oh wait I don't hear you yet Pablo sorry oh I was mute hello um it was a great experience to go through the document and review it and reread it and thinking about it actually it's a living document which is super exciting and I keep taking notes and and hopefully uh act on it on the future and uh and it's been a super uh experience to to find so many people with the same some interest same interest um some values that I really think are going to be the basis of organizations on the future so I think we are doing something very cool here cool all right well let's just walk through some of the sections in the 25 minutes that we have um again the paper is being shown on the stream but but Dean and probably don't see it so we kicked off with a preamble we wanted to introduce active inferences briefly as we could and point towards the fundamental first principles and physics-based nature of and initially Point towards its integrative capacity in principle and then in the second paragraph there's about 104 citations in this paragraph and that's where it's really only a sampling of the domains 21 domains here so it's a place where we can point to where is active inference being applied or How is active inference being applied or how is it different or better than another approach in a given setting which isn't to say that it does everything in all settings for all people at all times and all of that but this is to say that here's where one can look to find out more and even just building this and bringing it all together we we see so many opportunities for epistemic resources in the ecosystem and having real-time literature analyzes and all these kinds of services that we want to provide starting on page five we turn to The Institute and we

write as of August 2023 the active inference Institute is a registered non-profit organization tasked with identifying establishing and managing the sustainable implementation of administrative and governance functions to give components of the active inference ecosystem coherent forms and reliable channels of communication two publishing and Licensing protocols that establish open and fair use and effective dissemination of community products within and Beyond the ecosystem three services at the scale of individual humans and the community at large so that stability is protected while risk and uncertainty are minimized within the ecosystem and four organization and operation of cyber and cognitive security systems that ensure productivity inclusivity accessibility and safety in discourse in collaboration figure one gives a graphical representation just one of how we are now and for those who want to dig more check out the 2022 active inference textbook where you'll also find a golden ring and we kind of took that graphical Motif and ran with it had a lot of fun and a lot of interpretations there so just here looking at figure one Dean or probably just want to add anything yeah Dean so one thing about the building of this document um with a whole bunch of authors it's often not made explicit but there are two aspects to learning about yourself one is to actually throw some stuff down on paper because you believe certain things need to be included and then the second part is the consolidation process and I think one of the most interesting consolidation processes was this graphic because I think what what it does is it tries to condense as as much as possible and yet still remain coherent around what the relational aspects that we are trying to make available are as opposed to this is what we've captured now look look what our graphic designer came up with so if you've had a chance to kind of look at this and really analyze it and think about the layers and think about okay so what's the background here what's the foreground does this representation allow for a swapping can we have things that appear to be background or you know fundamental basis stuff that could actually be things that we are addressing in the moment versus those things which we look at as particulars that in fact are generalities so again it's one of those things I I don't know Daniel probably knows whether it's a where's Waldo's thing or you know some sort of artist that they came up with some strange Twist on things but but the bottom line is spend a few minutes have a look because you'll want to go through your own building of what these relationships are and you'll also want to be able to consolidate if you I wasn't there when this representation was built but I do know that I want to be able to both look at it do a comparative analysis to what I think the Institute should be and then be able to consolidate awesome yeah just a few notes on this and to give a visual overview since it may be clearer to

see this then go through the pros that essentially describes it as much later on the Golden Ring represents the active inference ecosystem and I think we could talk about golden rings for a long time to come but coming in from the ecosystem on the bottom bottom for the reader not from the other side of the table are the projects and all the projects are participatory Endeavors they're happening behind on and in front of the golden ring depending on how you look at it and that is the appendix and all of those different projects that we talked about and all of the iterations and new threads that I know that we're going to see in an hour and I know that we're going to see in a month and in a year in a century and those projects are being supported through to units kind of like a department but an organizational unit one centered on education and one centered on research the education unit is called eduactive and the Research Unit is called re-inference so right there you kind of have active inference in case that wasn't clear coming from The Institute scale supporting the unit scale are a variety of functions such as funding and Partnerships Communications and hierarchical processes such as the construction of enabling systems and at The Institute scale other than administrative functions we have the officers the day-to-day operators of the work the board of directors which is the formal governing body and the scientific Advisory Board where we really want to call out broadly and have people who may have or may have not thought that they would find themselves in a scientific Advisory board for example in 2024 this is your invitation and we really want to have a vibrant scientific Advisory Board who serves liaison type roles with different sectors and just gives periodic super valuable High leverage insight and then I guess depending on how you think about it as all things we have just a few terms they don't just represent bricks in the but it's more of like the the fibers we weave and some of these are already things that we do every day courses educational groups events live streams ontology research others of these are things that we know will be coming into place in the coming years and we know that the tapestry is just going to continue to expand so it's really exciting on page six we have another figure and this is just another way to represent some of the functions that we serve in the ecosystem this isn't the only final or a fixed representation of the functions that we have or the only way to slice and dice or any of that but just to connect some of the dots between the functions that we do and want to serve for the ecosystem so starting from the broadest let's just say where the listener is where you are right now is in the space of awareness and that happens through communication some people who become aware through a search or language model or a video suggestion or a friend or happenstance or synchronicity or however some fraction of the aware will pursue education and about education we have and will continue to produce

the materials documentation ontology courses live streams and so on education is produced and rests upon a comments it's an epistemic comments it's a verified information environment and that is really the space the wildlife preserve as I know Dean likes to refer to it not jokingly at all and underneath the commons is the support that does sit between a participatory Commons and a formal governing kernel that support in the coming years may go many directions but we know it's going to involve all of our favorite things like Hardware software information Behavior or narrative and more and then we have the regular Institute scale governance partnership sponsor and donor relationships such that we can first find a type of stability or sustainability and then more actively play a regenerative role in our ecosystem we want to dispense micro grants to people and we want to explore paid internships and be able to meet people where they are in their active inference Journey things have been happening to that end and through many means since 2021 on page six Pablo or Dean want to add any thoughts I'm super excited about the governance the education and the game and tools in an hour we're gonna see the first active inference game this is very fun on seven we describe our vision our values and our principles and just to read the top level on the values and principles they are active inference and exploration integrity and inclusivity Dynamic internal modeling anticipatory Behavior and continuous development on page a we talk about some priorities and challenge areas these are challenges across multiple scales they're challenges that individual people face in their journey to learn and apply in so many different backgrounds these are challenges of national interest these are challenges of global emergency as these challenges are educational in nature for example related to scientific literacy and Workforce Development and just general professionalization and Changing Times on the research side there are some priorities and challenge areas related to the fundamentals of cognitive science and grounding the cognitive science in physics getting that integrative first principles framework that we expect and prefer about information science and diverse intelligences things are faster and different than they have been and active inference is perfectly poised for sense making a decision making in that space user experience accessibility and socio-technical Design this is a major open question across different settings so it's not uniquely held to The Institute however we really do believe and really will commit to exploring new kinds of ergonomics and onboardings and experience so that we can come through and be how we want to be from the first time somebody enters the game where we can highlight accessibility meeting people in the language that they prefer to listen to and speak at the language in computer language or at the math level that reflects somebody's preference in that moment as

well as being able to use augmented and synthetic intelligence systems to ramp up and even push the frontier of what augmented people and other kinds of collective intelligence systems can do cyber and cognitive security these also are broader topics we've worked on these topics from an active inference perspective many have and we think that this is going to become increasingly relevant as things do become more and more challenging and uncertain in certain ways out there scaling the active inference ecosystem we've heard about this from the narrow and technical side we had estimates of the computational complexity of sophisticated inference about how many computer operations have to happen for a given calculation to play out and that's sometimes called scaling active inference like okay it took one minute on one gigabyte of data so how many minutes is it going to take on this computer with this much data though is a broader question and that's where we come to more of the social Fabric and the real sense of belonging that we all want to have and co-create and provide for others in the ecosystem and so this kind of scaling is really transdisciplinary it's really skin in the game it's as real as it gets and applying active inference some applications we've heard from today some applications are cited in the early pages of this paper and more and more come through every day and we support at The Institute various applications of active inference Dean what can you add in there why okay my beauty is off good so here's here's a here's something that never came up in the writing of the document but I think it's kind of interesting and I'm curious what you guys have to think about this 175 years ago an institution came and formed together it's called the Smithsonian Institute now I don't know if 175 years from this conversation today they'll be the equivalent of whatever the active inference Institute is 175 years forward but I think it's okay to start thinking in those terms it doesn't mean that we follow in their footsteps necessarily but I think it means that we have a long-term view of that the things that we write about ourselves now have some shelf life have some real sense of what we can't there are things that we can and we cannot change but if you look at this if you look at the front page the landing page of the Smithsonian Institute the first thing that they say is welcome which is what we say about the active difference Institute and then they say the Smithsonian institution is the world's largest museum Education and Research complex we are a community of learning and an opener of doors join us on a voyage of discovery so in three sentences they kind of sum themselves up what's the formula for not just existing for 175 years but being a force and I think again Daniel whatever you decide to highlight in the document today but I think we're really trying to do is say do the words that we put out now will they continue to resonate with people for decades to come and again I don't want to sound all hand wavy but I

want to hold up an example of a place that really did set itself up to be something that people could return to again and again and again and I don't think there's anything wrong with aspiring to having that kind of place that kind of respect and I think that the things that you're pointing to right now hopefully set the set the the you know the we're preparing properly so that's what I'm that's what I'm taking away at this point we know about that preparation measurement cycle from Chris Fields don't we we go on let's talk a little oh yeah please Pablo go ahead no no definitely I completely agree I've been thinking about that uh for a long time you you want to build an act uh for the best of the future and I think to reflect that on the paper it's super important so yeah definitely up to do that way ahead we describe some paths that we want to take but and and the history of the Institute we describe our Origins but I actually wanted to come to to figure three and four on page 11 to speak to a building one of the most amazing and encouraging things has been the all ages all time zone active build contribute mindset and earlier version of myself I might have thought like hmm people who do variational Bayesian methods and machine learning they're going to have no issue onboarding to active inference it's just going to be like as simple as saying yeah it's a variational Bayesian model with action so just signal processing on the inbound that's just control theory on the outbound but if you're familiar with variational bays and variational free energy this isn't changing it that much oh yeah we use a particular partition and it's related to the free energy principle but don't worry too much you'll get there I thought that that would be the path for so so many future analyzes will reveal what they do or don't reveal but what we found in our enacted work as it really played out that there was inclusion and unique contributions to conversations ranging from philosophy of science to fundamentals of statistics and meta science and the role of quantitative studies and social systems all these amazing topics and people who just scampered so rapidly through the prerequisite what could be seen as prerequisite from a disciplinary perspective was be was facilitated by the recognition that one was already in the space and in the way that they needed to be to be an act of inference and that was not written out before and even now the traces are barely recorded but it will become to be understood more fully and to your point Daniel I think especially what the pie chart shows is the actual Spectrum aspect of this so I know that right now that that maybe the neuroscience and the psychology piece take up a bigger portion but I think one of the things that we're trying to promote is the idea that um you can come from just about any background as opposed to Oh all all paths lead to like as you said an earlier version of you said oh okay well I can see where everything is gonna where the Confluence is going to no I think what this

District this demonstrates is that it's actually becoming more distributed the longer we we continue to to exist yeah this figure of course I'm super happy with everything that we were able to do but this figure is like almost a little bit like a um like a a paper Target that we're just going to blow out later because it was a fraction of only people who had formal affiliations listed which means only people for whom we had done full literature based polling up their information so it doesn't even represent the actual participation profile this is more like the invitation list for live streams we are going to see another long a long tail that covers the part that's not the long tail I believe the rest of the paper continues along possibilities challenges next steps more about the Institute organization various other pieces of information that potentially eager stakeholders or participants or donors or Liaisons or other people would like to learn more about it's there to engage with we want people to hop into our Discord and continue the conversation or email us or otherwise get involved it's why we called this Symposium enacting ecosystems of shared intelligence at the of course relevant suggestion of Carl friston because we really do want to enact it so if you feel like you're on the sideline waiting to jump in or you're on the beach and people are out there surfing the 100 foot wave then maybe it's time to get more involved so I will leave the closing word first to Dean then Pablo and then we'll go to raf's presentation well just real quick again if you have if you want to Consolidated document that speaks to the now but also points to a direction that we none of us really know is is certain but is also really exciting have a look at the doc Pablo um yeah I'm here uh to make sure that I act as one of those people of all backgrounds and perspectives that has taken advantage through one year of learning and acting on this institute and hopefully see you in one hour so thank you and have fun thanks awesome thank you thank you Dean thank you Pablo welcome RAF also author on the paper so how are you doing uh I'm good what about you pretty good other than this brief interval of white light I'm just hanging out hahaha fantastic so I'm trying to figure out here how I can uh I'm using a different uh slide software setting to make sure that I can pull this up in the right way so give me just a quick second yeah great cool so just just to um read off what's coming while you figure that um out so right now we have Ralph Kaufman and this presentation is going to be Gaia an active powered Network for planetary scale sense making following this 30 minute presentation we'll have available on embedded normativity that looks perfect that's perfect awesome all right thanks Ralph to you thank you all right I was going to try to see if I could blow this up to full screen but I I can't I got it I got it on the Stream oh it looks good on the stream go for it cool so I'm just pulling up my notes Here so uh let me just test that this moves did I just move the slide great cool all

right so uh as I said uh hi everyone it's a huge pleasure to be here and thank you so much to Daniel and everybody else at The Institute for organizing this I'm going to talk about what I think is the single most important application of active entrance not of in as a detriment to the others but it's sort of Upstream of everything else and that is how do the 8 billion of us humans make sense of what's going on in our planet and all scales and how do we use that understanding to survive as a species so this presentation here is a bit of a technical presentation and it has a lot of content it's going to move pretty fast and assume that you're familiar with some Concepts but I'm going to try to leave time for a q a at the end so first of all I'm going to talk about the motivation behind what we're doing uh in particular the concepts of the medic crisis and the third attractor then I'll give an overview of our Network architecture and the application so far and cover the many remaining challenges and last but not least I plan to convince you that this is the single most important thing you could ever be working on and I'm doing this because we need your help so that's a taster of what we have and uh so as I said let's jump right into it with the motivation and uh what greater motivation could there be um if you haven't heard of The Meta crisis uh it's a term that it's originally from Doctor Who but it was Borrowed by a thinker called Daniel schmeckenberger to talk about the total risk for Humanity and the biosphere that is posed by the combination of three factors the first being the increasing interconnectedness of everything from climate to food security to National Security to biodiversity and everything in between the second being the the risk associated with self-evolving technology including runaway artificial intelligence and the third being the increasing uh omnipresence of coordination failures personified as the the god of coordination failures and so the combination of these three factors as hypothesized by schmuckenberg who lead to either one of two attractors one of them being chaotic breakdown of uh all structure and the second being a reversal to oppressive authoritarian control and obviously both of those qualify as bad in my book I assume in yours as well so the question of course is can we actually find a third attractor that is positive Sun as opposed to negative sum and rather the question is can we design it so what I claim and what we as the the guy Consortium claim is that yes we can uh and we can design this attractor and it's the design goal is basically to build resilience and the stabilization into our biosocial economic system and to do this subject to multi-level system constraints like the applanetary boundaries desirability meaning it has to be something that people actually want so they will help it happen so it needs to preserve standards of living and so on and it needs to be feasible it needs to be achievable from our current initial conditions and of course there's a lot of design complicators in the mix uh some

of which uh partial observation there's lots of information asymmetry going on uh there's a lot of uncertainty about how the pieces of the world system work from a scientific perspective there's obvious computability constraints Associated to this and last but not least uh you can't really control what happens in most cases we can only create consensus recommendations and nudges uh easy right of course it's not easy but we think we can do it if we apply the the sum of the principles from cybernetics that you already know and apply them in a decentralized uh positive sum composable way so uh what we're talking about here is uh decentralized Network this implies hybrid AI human Network for planetary scale decision support and automation uh it features uh it recognizes that we will have human and artificial agents interacting by the billions in an open network its goal is to facilitate both consilience or model alignment and coherence or goal alignment at multiple levels or scales up to the global scale meaning the survival of the system as a whole it will need to feature built-in incentives and governance and it will need to be private privacy preserving intrinsically so uh the below list participants are participants founders of the of the Consortium I'm going to talk about it later but so this is the framing this is the motivation for what we're doing and uh and I'm gonna give a whirlwind tour of the architecture so far and I'm going to cover mostly what we've built uh the the version of this architecture that we've built so far uh thinking about this explicitly and then I'm going to talk about how this connects to other architectures that already exist in the world just very briefly so we've been uh framing this as uh an architecture for building decentralized digital twins uh which are local models of real world systems that run in and the road that such a digital Twins play is that they help us understand the costs and benefits of strategies and projects that happen in the real world and we in in our organization digital Gaia we started out caring primarily about agroecology so how do you actually make better decisions uh meaning better recommendations negotiations valuations investments into a form into a forest into uh uh bio light landscape to make it uh better to make it more likely to to survive and thrive so uh as a practical implementation uh it needs to tackle several real world challenges first of all uh it needs to handle distributed non-iid data uh data sparse it's heterogeneous it's private it's potentially unreliable uh it needs to enable localized collaborative forecasting and planning personalized planning and privacy preserving it needs to uh accept the fact that it won't be able to directly control but only recommend it nudge as we discussed and the the outline of what we came up with is this concept of the guy Network which is a mesh of AI agents called natural entities or ants for short that act as proxies for Real World Systems um each and runs on an engine called penguorn which it uses to inquire learn plan and allocate resources and

ants communicate in a language called the guy protocol they they use this language to independently interact with their environments and each other and if we do this right I'll show a bit more of this later but if we do this right the guy networks Behavior approximates a single composite agent that handles uh that interacts with and couples with its global environment just to give you an overview of how this works behind under the hood the fatron engine is really the heart of our architecture and the core Loop uh that is mentioned here is a typical active inference Loop but uh to make this work in very heterogeneous and distributed contexts there's lots of work involved so just some highlights that I think are going to be particularly interesting for this crowd uh first of all so as I said this first the same core engine needs to be able to work for very heterogeneous contexts so each agent has access to a library of generative models and these models are specified using a python based program probabilistic programming DSL and they declare what context they are appropriate for using a shared ontology uh using that ontology fangorn can then select the most appropriate model to use online given the configurations like project configurations and the kinds of observational data available so using the model using model that's appropriate for uh for a form or for a horse what kind of form what kind of forest what kind of ecosystem what kind of of you know their context energy projects um I'm gonna get into the examples right now because there's more more on the applications later on so uh we end up with a network of heterogeneous agents but that's okay they actually form a single model in a formal sense because they're all connected hierarchically uh by a system of of hyper priors and uh this means that Bangor needs to support inference of generic hierarchical generative models and we actually do this automatically using approximate posteriors with structural constraints um and so we have this Global model that connects all the nodes and describes General facts that apply over multiple contexts and this means the whole network needs to jointly perform inference over this Global model using a peer-to-peer or Federated algorithm so initially the way this works initially uh the network sets the global posterior to be the same as the global prior each node then independently performs local optimization of the variation of free energy and then passes the parameters of the new Global solution to the peers in the network where they use it as a hyper prior and this works over a broad range of topologies and under pretty General constraints and by the way I'd love to I'd like to mention that uh there's a lot of overlap and interaction between the stuff that I'm talking about here and some of the other presentations that you're seeing today and I I wasn't able to reference them explicitly in this talk but I I'm sure you guys are going to see the connections cool so um what it turns out is that the shared parameters that

represent our Global model are effectively stored by the network stored in a Federated parameter store meaning that each node stores their own local parameters their private stuff about their own local context and multiple nodes store copies of the global parameters and this architecture because it supports heterogeneous uh local models it actually allows us to introduce another kind of nodes which we call ant mood nodes uh these are nodes that aggregate scientific and empirical information using a meta-analysis model and uh we actually uh are able to use that meta-analysis model to constraint or constrain our project level models or ants using quote-unquote imported knowledge that wasn't directly observed by them which is left to right in this picture but this actually also goes two ways so empirical findings from the projects uh what actually works on the ground for this particular project also informs the end moves but serious which is the right to the left and so we actually have a complete uh type of Science and empirics all implemented using hierarchical active entrance so finally uh just uh among the global parameters that Network learns by Federated inference are the precisions associated with data coming from the different data providers and this gives each node a way to consume data without rustic ground truth and basically to learn together which sources they can rely on for what uh this is just uh some of more core features we're working on integrating many other capabilities into this architecture not going to go through all of this uh but basically what what we're finding is that we have all the building blocks to turn this into a full feature decision Network and complete with its own internal knowledge economy meaning that contributors uh contributions get attributed and rewarded in proportion to the to the free energy reduction that they afford cool so uh jumping into applications uh as I mentioned our main focus as an organization organization has been on modeling agroecological projects like regenerative agriculture and agroforestry and this exercise is all the constraints that we've talked about so far so uh again I'm not going to go into full detail but just to give an overview of what is the anatomy of uh generic model for a form of forest so we start at the bottom right with observables that are things that we can measure about that ecosystem about that that local context like the plant counts and size vegetation indices from satellite data and so on these are are linked to latent states which form a non-linear dynamical system that's parameterized by uh the policy the the the things that modulate the the latent States uh like agricultural interventions and practices we can think of the timing of plant planting and harvesting what kinds of crops or trees get planted what products like pesticides get applied or not and more uh as well as covariates so external events like weather physical risks Wildfire drought and so on um and so there's uh some scientific and measurement parameters uh that that go into this as

well things like uh the effect sizes of the various actions and covariates and these local parameters are conditionally dependent on a set of global parameters which as we discussed they they link information between different ants and different sources of knowledge um uh and uh so these same Global parameters are used in the meta-analysis model and they actually established the link between what happened what is found to happen on the ground and what is reported in the literature um and uh so I'm not going to go into this because I I think I'm gonna uh run a bit long but basically the concept is is that we have these project nodes that correspond to to many forms and they're each on Independent nodes in the network they're also connected to this common meta-analysis model and uh together all these uh together these nodes they incorporate academic studies expert knowledge and uh locally relevant context they did so privately so only processing the local data locally and uh through this parameter sharing they actually jointly uh and iteratively estimate the the global posterior uh in a cycle forever for as long as there is new data about the project as long as there's new studies and new expert knowledges so so all this is happening basically in an online way and again the topology can vary but the principles remain the same so just to see you show a little bit about how uh this works this is just two epochs of the the posterior the global posters and this is just location and scale four four two uh parameters uh the slope and intercept of the relationship between uh uh vegetation index and the tree biomass in a one given Hector or forest and as you can see this this converges pretty pretty quickly it's comparable to to local local lowly methods um and just to highlight how this can be used to to uh also the limits of how this can be used to estimate uh Source reliability uh here's a really successful example at the left is pretty pretty straightforward to use this to estimate reliability of uh of uh satellite data and for other cases like very local very uh context specific data you actually need more data or more data from more independent sources that we had in this demo here um uh another very useful use case of this is as we discussed we don't have direct control over what happens at a form we can only recommend and nudge often we also like we get there's a node gets suit up and that gets stood up for an existing project that has already happened and we want to find out uh what actions were taken in the past and the same uh the same generative model can be used to estimate uh what happened in the past in the example it infers uh when planting started and when harvests started for for different fields um so uh bringing all together one of the applications that we've been using we're really working on it actually uses all those Machinery together to validate claims about projects performance and to estimate future performance we've deployed this uh in real world projects uh this one this is the examples here and seeing

this photo are by uh are relevant to agroforest reforms in Colombia and uh we found that this actually helps the project developers and their funders to come to a shared understanding of how successful the project is how much it's worth in terms of its impact and its fundability whether it should be scaled up or or needs to change course uh what is the quality of the strategy versus other potential strategies for natural regeneration and so on so uh it's actually useful on the ground for decision making in this context cool so finally uh I want to talk about the end State and uh going back to our original motivation of solving everything I forgive you for your thinking this is nice but hey we're we've been kind of in this uh downward spiral towards the meta crisis for a long time and what you're telling me doesn't even come close to solving everything and you're right uh there's a lot of movie pieces that need to be built and connected but here's the good news um we can do this there's no mystery here and we don't need Perfection I love this quote from Edward Fullbrook that says that to define a policy we don't need an exact empirical measures or optimality if one jumps from an airplane it may be nice to have an altimeter but what one really needs is a parachute so there is uh only so much that we can get in terms of diminishing returns what we should be really focusing on is creating and refining Solutions everywhere and connecting them everywhere um and you might be thinking about how this is actually scale to Global goals and how does this actually get used uh for uh scalably uh in building into the the fabric of decisions in our in our society uh so about scaling to Global goals uh I will again won't be able to get into the details but maybe Conor mentioned this in his presentation earlier uh we have actually some proof of Concepts that we can actually build active agents that that are made of active and active agents so we've shown in this paper here that a collective of interacting acting phasions uh is able to perform approximate inference at The Ensemble level and that this further that this happens through a specific mechanisms that are mediated by uh by capabilities like theory of mind and goal alignment they they work together to improve Collective performance by eliminating ambiguity and through actively exploiting diversity um so that's one and the other question is uh how do we actually uh build this kind of stuff into not just these uh ad hoc decisions but into really the fabric of decision making and that that gives our global economic system so uh talking about here uh work that I did with Casper has this is a game that highlights the ability of ants just like the ones that we just described to infer true System state and use it to credibly incentivize long-term positive outcomes in uh an externality happy scenario uh this team uh we were able to show that it works even with adversarial bot strategies uh and even with high uncertainty and intentional misinformation so you can actually get these ends to to work with

budgets and incentives to drive decision making that internalized sexualities and uh and compensates for occlusion and uh uh generates long-term positive outcomes that's not all there are other Solutions uh uh that have been around for hundred and hours that apply similar principles to Global decision making problems in the real world at scale this is one case study from our partners at cognizant that they developed for the cool epidemic and this is uh not using active influence formalism this is a combination of traditional recurrent neural networks for prediction and evolutionary optimization for prescription and it's actually able to achieve some pretty amazing uh stuff it discovers a Pareto front between uh prevention of cases and uh economic cost it's uh it's also a showcase for hybrid human AI intelligence they did an initial version of this model then they launched the next prize they got a bunch of contributions for other models and then they did a meta model uh that discovered a better burrito front and uh even better even better this tool has actually been used to advise policy in the real world so it's actually very exciting that uh we're working with them and connecting the dots because we need to connect the dots we need to bring it all together this uh this illustration here is just uh I think a subset of the uh effects of the of the climate change crisis which is just one aspect of the meta crisis on on humans on our society and the economy and uh so we actually need to be able to deploy solutions that work across many different domains and to connect these dots uh this means two things uh the first is interoperability aligning decisions across context so that we don't get unintended consequences and second is reusability so being able to to transport and transfer uh learning and structure across context so for instance using that evolutionary approach from the from the coveted example in in our agricultural context or in many other possible contexts and this takes a form of libraries of components apis and so forth so that's what we're working on so we're launching this Consortium we just put up the website a few minutes ago and we have the goal of having a minimal but functionally complete implementation by the end of next year uh and this is an open project we are working on this at our organization called digital guy but we're also launching this open Consortium that welcomes uh contributors from anywhere both active entrance experts and non-experts and uh we need a lot of contributors we need all of you because this is uh the most important thing that we could ever be working on together and we need your help to tackle the many remaining challenges so if you want to help us build the planetary brain please go to GaiaConsortium.org and or reach out to me you know where to find me uh and together we can design build and learn what it takes to achieve planetarium generation thank you thanks for the great presentation I'll uh post the link to Gaia Consortium into the YouTube live chat but let

me get a few quick questions in so here we go first uh question this is from uh Marco Lynn I agree with the critical importance of such a project how will you deal with the challenges of scaling given the dependence on real world entities like humans for dependence on real world entities like humans for major meta crisis domains such as geopolitics socio-cultural fragmentation and other particularly hard sense making domains uh first of all hi Marco yeah thanks for the question and uh I 100 agreed that uh I'm not claiming that this is sufficient I'm just claiming that it's necessary uh we do have a lot of energy that that a lot of ideas and a lot of energy that we want to put into exactly what to describe how to uh change the uh how to weave what we're doing into the existing incentive Landscapes whether they're Market incentives or policy incentives But ultimately this needs to be uh an enabler this needs to meet people where they are and we need to be cognizant that change is not going to happen overnight at least not right now right typically these things take uh take a while to mature and then they they snap into place and we're already finding some of this happening in some of the domains that we're working on where uh nothing happened for a very long time and then bam hockey stick and we believe that the same uh can happen in and pretty much anywhere so we're very optimistic great what are currently the greatest challenges for Gaia as an organization I love that kind of ambiguous question like is that the guy a Consortium as an organization or is it but let's let's hear both what are the greatest challenges or frictions for the Consortium and for the bigger picture too also uh I I'm gonna I'm not gonna talk about the the technical stuff for a change I'm gonna talk about uh the the more Mana stuff it's actually very related to what uh your uh previous presentation Daniel about that paper uh about the Institute talked about and it's really related to to uh building out this common what we were calling the common cognitive kernel for for sense making and decision making and uh relatedly uh breaking through this uh this uh culture of sort of Suspicion and uh surfboards and getting um you know State people State trying to stake their their claim to something so we are and always have been uh extremely open but it's it's kind of hard anyway to build trust so we are were aware that it's going to take a while to build to build this trust with everybody and uh and I think the only answer to that is really to be as participatory and uh and transparent as possible also uh being humble and knowing where where we haven't we don't have the answer and uh asking people to help us awesome any closing words or thoughts well if there are no further questions I uh again I I thank you all for for paying attention and I'm going to be reaching out a lot more you're going to hear a lot more about this Consortium as I said it was just formed you just put up a website in time for this presentation and by the way uh thanks so

much to to Mahi and uh and of course Steve and everybody else who's been uh part of forming and uh and getting this to getting this ball to start rolling um we are um you know very early stage very formative and still uh very quiet so we definitely are we lost I I just gave you like my perspective on on a bunch of these things the the shape of it that that has come from from or work on this so far and even not all of it just piece that we could cover in 20 minutes or so uh but we are we're very plural we want to uh engage with you on uh on different paths forward different opportunities to to build things experiments to connect the dots to learn together and yeah ultimately like we we really just want to uh provide a platform a convening space a protocol uh uh in the classic sense of New World the word of a you know a shared language a shared means of communication a shared means of understanding to find together what we need to do to to actually build this third attractor call Gaia trim tab thank you exactly thanks so much Daniel and see you guys soon bye bye what a cool presentation next up welcome avail we have available with embedded normativity so thanks aval to you hello can you hear me yep we'll do and can you see the slides um I only see the top left of the slide you only see this the top left yeah just yeah try again thank you okay Yep looks great thank you okay good so I am uh I'm interested in a multi-scale collective organization this is why I came talk to in France to begin with and the core question I've been asking is um basically when we look at human's organization there is great Evolution there is a challenge through time of the specific patterns that are enacted and this is very hard to account for and I have been trying to pin down more and more specifically what exactly is going on and how to model it so this specific work is a work I wrote in the context of what my PhD of course which is funded by The Escape project a project in a cognitive archeology that's interesting interested sorry in how the culture the archaeological landscape shape cognition and vice versa the title of it is onomatic normativity an active inference of urgency an active influence account of agency being flesh because you will see very quickly why so first we will go through the existing work on ankle recognition in the context of acting in France then we'll try to expose the concept of embedded motivity which we introduce here and what it means and then we'll look at the more semiotic question around the externalization and utilization of normal activity in cultural landscapes so first encrypted cognition inactive so we interact in France I will go quickly over it I still need to do it because there are Lookers that are not familiar so active in front say that uh agent continuously predicts the specific queue sensory cues they will see and they predict the action or policy they will not take and they basically enact an ongoing stream of expectation that they reconstruct given you know a coming signal so this is best accounting

for in this schema which shows the agent as the miracle system which engage through environments through action and cessations and all of this um specific domains they minimize a specific function that is called free energy and whose minimization basically means there is a prerogative process going on and a core property effective in France is that it dissolves agency as we typically understand it we don't have agents that have like internal monologue that drives them to take decision we have a constitution dissipation of what's going on and whatever an agent anticipate itself to do it will do and so this goes for computationally satisfying but very definitional notion of what is going on with agency so the question then comes uh what is the relation between an agent and the cognitive landscape in which it evolves uh I will present very rapidly again the framework of skill and functionality appear nearby Bruno's and which says that skill agents which we understand as agents that understand what is going on around them they will basically experience the world as the landscape of affordances which is opportunities for actions and they will the recognition pretty much reduces to the experience of the Sunscape and adaptive engagement with it which basically maximize the grip the agent as on the system and essentially for reasons we will not go over because it's math the very uh existence experience of this landscape it is equivalent to the existence of constraints of our attention the attention that the agent gives to thing which uh translates the emerald expectation that are at least two times about it but the more expectation that the agent enacts and so this becomes very interesting when we you know cross it with the existence of enculturation in humans because the basic experience we have of affordances of what we can do it is uh shapes culturalities shaped by social interaction when we learn uh with other humans about what we do what we can do and what we can do and this means that the basic human experience it's thoroughly encultured it's embedded in what ROM said called a control landscape landscape of cultural fondensis and it is embedded in cultural and material [Music] regimes of attention so ways to pay attention to things and this is a figure that comes I believe from this year on thinking throughout the minds where it shows basically and how the actual installation cycle when we have the Civil agent they build a common environment that has epistemic cues on how the world is like and when we are incarcerated to pay attention to things in a certain manner that makes it so we perceive and enact this this landscape and so the uncultured constraint of all the flow of attention because of the what we saw just before the deflationary account of agency it means that the very expensive landscape cultural landscape that is collectively enacted it effectively integrates intentionality and it effectively uh integrates normal activity in a way that Clash is strongly with classical account of both so

given this state of the art let us look at what I mean specifically about embedding normativity so a marine motivity is a kind of alternativity I think you have you have got it so normal activity it means basically the criterions by which we judge things to be good and bad in the context of biological system they will look for specific I don't know chemical concentration and I will avoid other this is the phenomenon of chemotaxis this is pretty basic and this is a form of normativity but for humans we will typically think of you know laws cultural norms um there's really shared way to judge Behavior and again at least in the context of flow and choice Theory which because you think there is an agent he has access to information and we'll think Hmm what should I do and He will judge uh rationally about what is good and what is bad based on a collection of values which are internal to the agent this is not how it works what I call embodied motivity is by contrast regimes of normal activity where it is the uh the norms and value that guide the agent's Behavior it is embedded in the material and cultural environment so basically the agent projects a specific set of norms and value in his very experienced princess environment and it will perceive the locus of normativity as the environment and not as a form of internal judgment let us look at what can exactly how amazing motivity work it will be more concrete so first we can materially constrain people to do things so for example there are roads you can go off-roads but it's harder and you can't really do that if you're a chariots so when you have a specific pattern of roads which is constructed by cities uh you were pretty much constrained to follow the roads materially and for this we will talk of structural embedding and there is another regimes interactivity where we build on environment that will basically queue you suggest you into uh adopting specific norms and specific values for example Escape things that vertical patterns in archaeological culture they are somehow embedded within uh what we call vertical Norms so hierarchy and certification in Social relationship and for this specific form of mudding we control curve semantic compelling I'm what is must be very clear is that we don't have structural embedding on one side and cementing embedding you know there are basically two mechanism complementary mechanisms in which uh nominative sorry embedded multivity occurs if we look at the city so this is a impression of Sumerian culture city states we have much organization that prompts specific kind of behavior so for example we have a bridge and this bridge allow you constrain you to go through the river in a specific way if you are on foot but this particular constraints it is perceivable so by using the bridge you will also see that you can navigate the world in this specific way so you don't have aspiration between semantic and structural embedding we just have two ways in which um constraints of our behavior that are

embedded in the environment work something that is a core to the account of a medical activity is again that it is not internal but we need to go into more detail about what we mean by not being internal because of course we're talking about agency agents constrain what engines do it's not outside uh what it is is and Marine motivity is the origin of the motivity where the norms and value they stem from the experience of foreigners for example if I see a fire I will not think about hey could I put my hand in it and get my hand burn I think pretty painful I won't do that if I even think about putting my hand in the fire I will immediately feel bad the normativity it's embedded in my experience of the world I feel like the fire or more specifically the possibility of putting my hand in the fire is intrinsically bad I do not feel there is a possibility that is neutral and then I judge it to be bad but uh of course this uh this judgment does not occur outside even if I feel like it's about my affordances it emerges from the arrangement I have with the material and cultural niche in which I find myself and if we want to to untangle what this mean precisely we will have to go into more detail into the what I call the externalization and alization of anonymity so let us look at something very simple how normativity work for embodied agents so I'm read agents they do things with hopefully in an Niche that is material and this Niche will be marked by the activity of my agent and these marks they can then act as constraints either through structural augmentating embedding in the age environment because like basically signify that in the past I wanted to do that maybe doing that is you know good for example uh we have Centuries with I don't know if century is a English word but we have path if uh animals that have legs this is actually important work uh sorry work on a specific path they will compact the Earth and it will become improper to grow anything even grass and you will have a path that will emerge that are pretty it's pretty visible and I tell you hey not only this Earth is Compact and allow you to walk on its like without effort but also this has been widely judged to be um to be a Goods uh Outlook outline for navigation in this specific environment so the very experience of a path basically invites you it affords you perceive the affordance of walking through the path and this is something that again can result from your own activity that can be written into development by your own activity and solve this uh that's the question of how okay there exists embedded normativity but how do we embed normativity what is the activity of embedding motivity like and the word I will use for that is externalization we perceive the world because we externalize uh specific causes of sensation into an imagine uh objective external world so when we see the path we see a path we do not see compact Earth that we then compute to mean that other things have worked on it we just see hey there is a path and this is because we have

externalized a specific cognitive kind which is the path and this is an argument that several people named here are Lennon Steinler which if I remember well um worked on the um the production of devices that help blind people see things through I believe an array of Repertory device on their shoulder so the core argument they make is that if you give people that they will start seeing things they will start pursuing things as being before them rather than having Liberation on the shoulder and then thinking about oh what does it mean and so the basic argument is that perception is just um picture of what cause we infer what is the structure we infer of environment given the existence of a specific coupling device that relates action to sensation and vice versa and this critically relies on the internalization of specific we could say a lot of things but I will say priors specific prior is not an interception so missing the path as a path relies on me having the prior knowledge that there is such a thing as past and I should be expecting to see them around and so you have um duality of externalization and internalization in the production of a marine motivity I can only see Norms out there if I've internalized integrated or specific operators of perception that entails that they are indeed those specific Norms that are embedded in those specific references and so this uh produce a marginal question of symmetries basically of information Theory how do reading and writing motivity works uh so my next 10 lies I treat the world as having specific natural categories that I pretty much imagine and I can use them because I can adjust them as an extended memory that allow me to read and write normative cues uh this can be um you know letters uh pure signs but as far as I'm concern here all signs are at least implicitly Innovative there about how you should interpret the design and this means that we can experience a specific landscape or valencies and we can we can write things out there we can write normative cues biobehavioral that will then constrain for the behavior but nothing tells us that we can that we can communicate through this the communication it works because of the internalization of prayers if a given community of agents are vitalized a specific set of prayers that is a set of pairs that enable externalization of the narrative cues then only it can use the environment as a way to two rights and reads normative cues as a embedding normativity thing and this communication it can be read in terms of information theory of course it can be read in terms of semantics but what is uh critical to analyze that the process of internalization it is there there is no um trust system there is no private key that allows you to be sure that the way you internalize Norms is very specifically uh the way in which the amylene motivity was encouraged to begin with there is a loss or there is ambiguity or there is I'd rather say it like this an active the march of reconstruction of given prayers by the agents and so this gap between the intent of other

Communications and my intents my understanding of what I meant by their acting active engagement with the world it builds a tiny Gap in information that is somehow enough to drive the opinion evolution of cultural landscape and this is something that I do not think is a marriage inactive in France and I think is of major interest to research to the research at least in the evolution of opponent evolution of crucial landscape so uh to sum up things uh I just presented the paper that is called animate motivity and that speaks and surprisingly enough of embedded normativity uh first major point is that effective inference is correct then um our notion of agencies dissolved essentially in the notion that we can perceive and enact cognitive landscape if you want and given that this is the case we will experience the norms and value that guide our Behavior as a property of you know the cognitive landscape experience the affordance that we can perceive and this specific regime of normativity I call embedded normativity and the third thing is that you have a critical thing going on any other symmetrically constraining thing going on in the externalization and utilization of regimes of embelline motivity because those two things they respectively enable nomadin activity to exist to have causal power and the enable also the operating Devolution of the Norms that are embedded in a given landscape in a given measure landscape and therefore the operating evolution of the assaded cultural or cognitive landscape so thank you I will enroll my references slides in case someone has the really needs to know what the reference season is ready to move around the video for this and do you have questions awesome thank you avail while people are typing any questions could you maybe just give a little introduction to the active inference for the social sciences course like what brought you to want to facilitate this effort and where are we at and where do you want it to go what do you want it to be for people and then continue good point thank you uh the active info Institute organizes uh and keros research organize a course on the on active inference in the social science that is essentially in explaining the um basic the March the basic hypothesis and the consequences of the cultural landscape formalism that I hinted at slightly in this presentation the goal of the course is to give a basic ontology that is on the interface between complexity sense let us say physics written large cognitive Science Central science to afford interdisciplinary collaboration in the in the social science and in the study of the evolution of human societies and I think it is cool but then I asked because I have a nice date some maybe you should ask Daniel which is absolutely embiased whether the course is cool I I've been enjoying it a lot uh we just finished last week with my section on Collective Behavior first we had the introduction lecture from you then Ben why on the basics of being an active agent and we have Lorena and mouse

sections upcoming as well as your more dedicated session yeah it's been a really fun uh uh experience and it's going to provide a really useful missing piece in the ecosystem I guess just one question I have how do we take some of these topics you're bringing in about the embedded normativities and think about them digitally like are we talking about conceptual embeddedness in information environments and or are we talking about the ergonomics of sitting at our desk or how do we bring that kind of pleasure stigmergy on the grounds into the digital are we talking about something that's embodied and embedded in the information space or are we still talking about bodies so if we talk about humans we talk about bodies with the brain needs that do things in a material environments but your question is uh I'm not sure on the studies how can we bring these specific Concepts in the study of digital filamental I like Twitter internets yeah is that it yeah okay so Europe basically is the whole study on the uh affordances you have a so the recipient of design has incorporated a while ago the National Conference in its structure and most of the study in the digital design center learn what uh platforms are for or do not afford and we have a richer stance when you take into account not what you can do with a specific interface but what the specific interface invites you to do given a set of cultural it can be self learning but given a set of power that you share with with the um platform in a way that is a formal but indirect and so yes I would look at it in terms of in user interface design I think this concept specifically is useful for for interface design then you will have implicitly the question of um the floor that platforms such as I don't know uh Twitter Facebook um how specific structural environments bring specific Collective dynamics that might be a bit um richer not richer specifically but that might go in a direction that the user interface question do not go into and that can still be modeled in terms of constraints and therefore in terms of environmentivity is that on point we'll go to a question yeah that's interesting about what the interface does and then what it invites it might invite things in order for that invitation to be fulfilled it has to actually provide that possibility but it might invite things that that aren't relationally possible like there might be a wall which invites climbing to a given Observer yet they cannot climb it and then on the other side a capacity that isn't invited and that is the the adjacent possible of tool use where something that was being used for this purpose it's like all of a sudden it just fell out of someone's hand they grasped it a different way it invited a different engagement and then that can become re-entrenched and that points to that sequential like making and breaking of new spaces that your work has featured over the last few years exactly this work has been developed with uh while thinking about tool use while thinking about the cognitive archeology we're thinking about the

experience of cities the way urbanism from specific behavior uh but um if I can get back on the internet thing I'd like to make it more specific so the interface design as far as I know is focused on interface let's see this that is not surprising but in doing so it is focused on specific foreign what you can do you have another layer in uh ecological psychology with his the layer of solicitation so for example Twitter there's something very specific which is when you log in it shows you it shows you the number of notification you have then it removes it and you have to wait to know which were those certification and you have to wait no once again not one minute but a random amount of time so this is uh what they call dark patterns they are patterns in design that are explicitly designed to create addiction so this is a layer you can't afford in a France because uh from a foreign motivation in the way uh specific structural environment is perceived the simulacra semantics Maybe the meaning that given landscape has for an agent and even if you look at this you are still still looking at the platform in a way that is um is embedded from these specific patterns that specific patterns of things that people do on the platform if you have Twitter without the aggressive qualities and the um way to aggressively signal that you are in whatever it is is going on you don't have Twitter much of Mastodon maybe that has very similar affordances very very dissimilar solicitation so this is this is why you have to take a multi-scale account of things and when you take them kind of things you have to look at the semiotics basically Behavior how specific coordination occur at specific scales around specific uh communication specific message passing and once you look at how specific message become meaningful to specific agents you are you you have a problem basically because this is subjective this is constructed through the activity of the agent and this is uh whatever is the range of time you're looking at you will have the construction of new possibilities of adjacent possibles as you put it and yeah this is the this is the uh big question that I think we are missing at the moment in active in France awesome well thank you for this presentation thanks for the social sciences work and we're all looking forward to seeing how the projects play out thank you goodbye farewell level next presentation is with Pablo Fernandez machiara so welcome back Pablo hello akir okay I did take us on an adventure of curiosity yes I'm gonna start sharing my screen let me know if everything is all right should be yep cool so this is one of my favorite memes and probably most of you know it this reference the cave of platon and uh to me it was kind of a fine intuitive truth that now with active inference seems like we are closer to demonstrate that that how it is though the first time that I start uh act in and surf at The Institute uh I just this was my kind of my feeling or my face uh was uh looking into something that it was complex scary and look interesting so I went and

why I decided to go through it ambition a world where people is driving by stronger incentives that motivate them to reduce uncertainty in the systems they are part of and my Approach is through games which are gonna go later but I really on this talk want to encourage uh all kinds of persons or people that are curious and I want them to act and get involved in the community because you may find that uh for now most of the actors are scientists and very well on reputed people and you might think that you are not a part of it but uh very far away from that uh The Institute encouraged uh the opposite so I'm super glad to have been through this journey and I wanted to share a little bit of my background and then we go to the game here is the second and last meme uh the world you were raised uh to survive is no longer exist we are on a digital Revolution and with this digital Revolution uh it comes bunch of different things that are evolving and hopefully uh bring us to a better world um okay who I am I'm Pablo my name is Pablo but I'm more defined or well what best defined me is my action and my experiences I think uh role game plays um Sports and movies are super important and asakida was uh uh dyslexic and the system didn't feel my Necessities so I really hated it's cool but I love learning and it's something that I've been doing all my life and it's curious how a system that is designed to give you knowledge and to help you with the learning in my case it didn't work at all and and just that's a little uh starting point uh I believe in in the copy transformation and combine of things and on my uh adulthood uh everything sifted and I became a successful professional and I love it teams what they do for living is a product manager and what I do on my pet projects side projects call it as you want uh is related to uh gaming and gamifying uh organizations and experiences um uh nowadays I trust more my intuition which is something that it really was broken during my childhood and you will see a little bit more about that here are some things of my interest licenses uh like open science which is the case of the Institute philosophy and that's the first meme education governance family art um law of communication technology and nature and a bunch of other things that I will be happy to share any other on one of the questions that Daniel asked me for this talk is it was what was exciting for me uh uh about all this what right and for me I always when when I I give talks I I like to Sir or a view of the last hundred years and and the Industrial Revolution and how uh the the ones that were on this world uh before us did something right I'm sure that we all know uh big mistakes or big things that have uh fail or that could be uh better through iterations but on data it's it's very uh promising and and very good I I think is um inspiring to have uh these the basic education how it grows uh the vaccination what it did did for us uh democracy uh and how uh poverty drop drastically how

literacy it it grew and because all these uh things uh child mortality uh just went very low uh so basically the future of education is what makes me be excited how I ended on this ecosystem uh it's super random and I'm gonna explain it in hopefully two to five minutes I as a product manager I'm very interested on technology I heard about blockchains back in 2015 and I didn't understood anything at all and then back in 2017 I heard about a stos I didn't understand them I did not play that game but it was a racing a fun thing to consider which is digital assets and ownership of those digital assets uh for now we we did we didn't had uh a technology that could scale and that was good enough to ensure that the digital assets that you own uh were you yours and you had the freedom of transaction uh on them I I did understand that very fast and because I'm a game player and as a game player you always have your different assets and and do things with them and as well I I I have been collecting music and all kind of uh digital assets uh I bought uh this very famous collection of uh punks and sold them very fast before uh I Could Be A Millionaire but during 2021 I decided that I wanted to build something and then back then I went through this uh beautiful project that was the first art show on ethereum which one of the artists it was Daniel uh Friedman so I as a curious person and a person that has been working with artists I always like to know what is the driver of those artists uh what they do and and try to get in their walls to understand better and to get as much information as as I can before I decide to uh encourage or collect or or or just deep deep always uh deeper on on music or art for me is um very nice nice thing to do so I discovered uh these cards and education was one of them and I went through uh Flickr which is where it was the first at least what that I found a place where it was and the license it was a Creative Commons license uh of not commercial use but uh it it was an open license which I been uh a part of uh Creative Commons Community for a really very very long time since I was uh uh working with musicians and I think it's one of the best things that happened to the internet um this is these are some of the pragmatic uh uh things that uh the audience cares about hopefully we will have some of these on our games uh uh AI for sure is going to be there free energy principle is going to be one of the things that hopefully we help to minimize and to make it funnier games or happier games or on games that help organizations to be better uh the human factor uh it's super important for me an implementation an action I learned by doing and it's one of the values that the Institute encouraged more so uh I just jump on things and try to understand them through doing and sharing as I'm doing uh here are the epistemics and um Collective uh is one of them that um as well uh super interested um agents of course practical and curiosity which is the adventure that um uh I have the feeling that I'm being

involved all my my life and that I encourage always my my two girls to to be curious and to open things and look behind the doors and just be an active here are some values that I like to point with our exploration uh the culture of curiosity that I did mention a lot uh continuous learning uh because that experience on the school I think I I just keep going and keep going and it's something that sometimes um bad experiences could turn out in good habits uh learning by doing as I already told integrity and inclusivity honesty I couldn't tell you accountability a big one um trying to be professional on everything that you do um for me I started as a human being curious um people of all backgrounds and perspectives are welcome to them to The Institute um I could I could be very happy to be the link and to speak with uh someone that has doubts or if I can help I think I could help on the very early stage and and if you're not very technical person I think I could help there and and yeah and NASA we are learning by doing I would love to everyone that's uh that's on the stream uh I'm gonna share this link which is uh a new a new game that we are testing and I'm excited to to share it with with you uh it's the adventure of curiosity um here Daniel I'm gonna be and paste the link uh it would be awesome okay there's a tiny typo on your slides it just ends with a curiosity thank you but I'm gonna I'm gonna paste I'm gonna paste it into the YouTube live chat okay thank you uh yeah I told you um dyslexic I have all kind of titles throughout my life sorry about for that I always try to improve it uh let's okay so you sir the yeah I put it in the YouTube chat so people can head over there all right yeah could you tell us about this platform and this is the first active game um we uh started thinking on how to do a game a while ago and we decided to go with this uh picture that explains how it works actually it's simple and you have uh for pieces the internal States external estate and in the middle you have the sensory State and the active state here is the mission you have some instructions that you can read and would be fun if you go through them and you can go either way and here we are gonna go to the sensory state where you're gonna listen to a music where some notes you can sensor that are not in the famous song of for release you have to take notes you can go and you can see that we are on the Markov blanket which is so everything very delinian but painted and here is your active uh part of the game where you have to do something which is add the different notes that your professor music Professor is telling you don't miss because you probably on the external state are trying to play the music so you have to infer that you are missing some notes go the active state and start uh walking through the right node if I miss the note I just follow the form into the nothing and come back it into my internal state so I probably would have to go back and censor again and once I take the right actions and the right nodes I can go and solve the fossil

and the idea here is to have a tool a gaming tool to explain or so you can experience how this active inference on free energy works and once you go to the end of that Adventure which I'm not gonna finish because it's something that players should do you're gonna have a door where you have to put the sequence in low case of uh of what were the missing notes and it's gonna allow you to come inside the treasure room where you will have uh feedback feedback that you can send us and we will be very happy to have it and then if you have a wallet you will be able to have a digital asset if not we will add information to help people to go from web 2 to web 3 and to get those digital assets here to be developed would be future Adventures that I would be very happy to so this was pretty much the presentation hopefully we have play new players in the coming days hope you have fun and I really really encourage anyone to discover and play with all these ideas and thoughts and experiments and things that I have that gut feeling that are gonna revolutionize how we understand uh the world and how we can improving it by reducing uncertainty so thank you and I don't know Daniel if you have any comments or something in this last 10 minutes oh yeah let's we can definitely talk for a few minutes I'm I'm seeing many fun comments and questions in the chat so people can can post it yeah the gameplay is smoother on even a normal computer it just like it may look choppy on the live stream but it's quite a nice um playing experience like maybe just a little context like how did you get into building these worlds or what kind of tools are these and we obviously don't see them in today's mainstream educational and research offerings but like what could that even be to bring together this kind of immersion and role playing with education research um I uh uh um we I I I I've been playing role games on Roll games uh you have uh very uh diverse kind of players so I I was playing with uh guys like me like uh had a struggle through the school and guys that uh end up being uh researchers like yourself because they were very good and and in those environments and and and for me it it was like we are sharing a lot of cognitive uh um values and information and they are really having fun with me so uh that what what the Academia is telling me is not isn't is not what I'm I'm feeling in in the real in in outside of Academia so that that through the years uh it became more and more clear that was one of the drivers to have a very diverse kind of friends and and and that one of the most important things for me to learn it was having fun and to be a good professional I have to be enjoying what I'm doing and to enjoy what I do it has to have that gamified feeling of it and so in 2021 uh I I just went to try to explore that road uh because blockchains what what can do is have a persistent sea of of objects and attribute attributions to those objects and to those wallets that you can add and build on top of them is like having a huge uh database for a game

purpose and and then you figure out that a game is uh kind of our organization so you need rules you need to serve values you need to uh have like uh what we are building now which is the session which you agree with the players what's going to be about and how it's going to work and and we are uh building this um fantastic uh storytelling of how uh kind of world where it was in the uh early 20s compared to what's happening now in this imaginary world that's complementary to what's happening in real life and we are just exploring those areas for three years and once I got into the and to The Institute we started sharing these ideas you send us to do um system thinking course that it was super uh cool with we we explore and learn and these uh document that I can share it was very uh important for us to understand and we find out the necessity to be able to computerize what we are doing so we'll start focusing more on ontology on on on all the the Brick breaks that you need to to understand uh to build um the skeleton of uh something that could last for decades which is uh the idea and we are trying to build it so everyone can copy paste Fork it and do whatever they want with the game adapted to their own organizations and use it uh on their favor and and we are just exploring and having fun that's that's awesome and there's so many the so many pieces there it's true about the fun and the Curiosity those intrinsic motivations like being six hours deep into counting ants in the desert or 5 000 pipette pushes deep or whatever it is in the trenches and in The Last Mile and um it it can be easy to lose sight but when we lose sight and when we put out the fire and the spark it's a cold dark night and no career or status or extrinsic valuation is gonna provide like simply the joy and the the deeper meaning of like two people going on a little Adventure running around this space being inside active inference rather than active inference being like this high-speed train that's leaving anyone behind or you're on the outside and it's taking these twists and turns and you're getting thrown off it's like it is cool and air-conditioned in this space and people play and watch a lot of games where they wield violent arms and not even like saying whether that's good or bad it just it is a genre to want to have a camera either in a character's head first person perspective or a camera in a second or a third person perspective which you can switch between here and like those kinds of topics about like egocentric versus allocentric navigation and our ability to have a a perspective swap on ourself and be able to take that kind of like out of body um situational awareness in real spatial settings and just to be able to cruise through this space and it's like yeah there's no there's no right way to Boomerang from extra you can go from external to sensory you could go here or there it's it's a um it's an appetizer and uh a fertilization that like we've really just never seen okay yeah thank you very much Daniel well for this any last thoughts or comments no just

enjoy while we are here which is I think our mission all right thank you Pablo I'm sure everyone's gonna really enjoy this and let's also of course continue to um develop these games and continue to offer these on-rams and have times where we're meeting up in these spaces and playing around thank you bye farewell welcome Mao this will be the final of uh presentation sessions in the symposium this is Mal aberration with the presentation shared pretensions in active inference so thank you Mao to you and you're the first you're the first to join with an AI Steward slash Guardian awesome shall I share my screen yes Let's do let's do let's do Advanced just a second I'm looking for a window I think this will do okay so can you see the screen Yep looks good thank you awesome so unfortunately I imagine that you are seeing the entire screen and not just the presentation I apologize about that but I got it it's it's cropped I got it okay wonderful thank you welcome to the presentation on shared Pro tension under active inference we're going to cover a lot of topics um should I introduce myself a little bit or have you already done that awesome so my name is I'm a PhD student in cognitive Computing I work in active inference in the interaction between uh social sciences uh philosophy Consciousness science and artificial um we're going to start a presentation on potential protection and active inference so I'm going to define a lot of terms it's going to deal into philosophy terms that are very deep um and so let's get started if you will just give me one small second and we're back so shared retention is a term derived from phenomenology and it refers to the shared anticipation of future events among a group of individuals this philosophical study of experience and Consciousness becomes particularly interesting when we consider it in the context of active inference when we combine these two concepts we get a powerful tool for understanding how groups of individuals can coordinate their actions and work towards a common goal this is the main focus of the discussion today so imagine a group of people planning a trip they all share a common goal to have a successful and enjoyable trip the shared goal forms the basis of their shared protection they all anticipate future events related to the trip such as booking tickets packing their bags and visiting various attractions as they start to take actions towards their goal they're also engaging in active inference they're making predictions about what will happen next and then adjusting these predictions based on what actually happens so for example they might predict at a certain flight will be the cheapest based on their past experiences but when they actually check the prices they find that another flight is cheaper so they adjust their prediction and book the other flight this constant cycle of prediction and adjustment Guided by their shared protection helps the group coordinate their actions and make their trip a success so we're going to delve deeper into the concepts and

explore the role of past experiences known as retentions in shaping our current knowledge and future predictions explain let's explain first the concept of retentions retention is a key component of time Consciousness that retains the past trajectory of a temporal object it's the preservation of the content of a now passed High letic datum in our present consciousness a high letic datum refers to the raw uninterpreted Sensations that arise in our consciousness when we retain these data we are essentially keeping a record of past Sensations and experiences that can inform our present and our future actions inactive inference we can formalize retention by thinking of moments as binning of states which are bounds to a hierarchical state this way retention is seen as the bidding which becomes the current state informing the next expected state to further connect retention with active inference we can consider the way in which retention plays a role in updating an agent's belief and generative model the past experiences influence the current expectations and actions thus shaping the agent's understanding of the world and the future so let's consider a practical example to illustrate this concept if you had a good experience at a particular restaurant you might choose to go there again in the future your past experience influences your future decision in hustling temporal phenomenology we would think about much shorter time scale but if we extend this concept we can see a much larger structure whereing the retention is simply the past state it's like having a mental note that says this place is good and this is how an example of retention works in your everyday life they're not merely representation or memories but presentations of a temporally extended present which distinguishes them from Recollections this involves the presentation of some already constituted content in conscious experience retentions also play a crucial role in mapping past knowledge onto current beliefs contributing contributing to the development of shared narratives or goals that we will delve into in the rest of the presentation they can be former now points and the current awareness of the past and far sedimented retentions can be engraved in scripts or deep temporal priors or more fluid deontic cues this phenomenon corresponds to a specific application of message passing between agents in the present moment and in the sensorium retentions and protections may also emerge jointly within a group forming a collective understanding that transcends individual perspectives protentions as we saw refer to the anticipation of future events based on past experiences and current knowledge it's thus forward look it's the forward-looking counterpart to retention which is backward looking protection is the anticipation of future trajectories of a temporal object it's the process of looking ahead predicting what's to come based on what has been and what currently is so in active inference it could be formalized by considering the future states

that are expected based on the current state and past States pretension is thus the anticipation guiding the next action you can consider the way in which protention plays a role in updating an agent's beliefs and generative models influencing the actions and thus shaping the agent's understanding of the world ends the future so for instance if you see dark clouds in the sky you might anticipate that it's going to rain and decide to carry an umbrella this anticipation of future events is protention it's also going to determine whether or not you're wet when you go outside and therefore whether or not you expect to be wet your brain is constantly making predictions about what's going to happen next so let's put this in the context of temporal flow which ties together those concepts of Primal Impressions that we will see retention and retention together temporal flow is the structure that represents the continuous progression of time from the past through the present and into the future agents are constantly revising their generative models and they do this by integrating the observations which we could refer to here as Primal impressions this updating process is a balancing act between maintaining the predictive density of the agent's current beliefs and adapting to new information retentions and protection structure time for a given agent creating a framework that shapes their perception and anticipation different agents or systems will exhibit varying qualities and quantities of retention and protection which resembles something like Mike Levin's concept of cognitive light comes so given the presence of retention we can anticipate a potential Horizon not merely as an arbitrary expectation students are thus the point at which agents beliefs about the world are updated to match their current which leads to a more accurate understanding of the environment and allows for more effective goal-directed Primal impressions are the Fulfillment of a protention which is the time-bound Nexus of perception and existence is it is resolved sorry it is the result probability that redefines the local radiant it's in Primal impression that reality collapses the Primal impression is thus mode modeled as the essential Bayesian cognitive moment it is the juxtaposition of Prior beliefs or retentions with ongoing sensory information in an unfolding future oriented present retention so now we have a description of temporal thickness from the nomenclature which is compatible with generative modeling in active reference agents for hustle this flow of conscious experience is primarily made up of Sensations that dwell up in a raw uninterpreted form that we called earlier highlighting data so this data is the now phase of a temporal object it is formatted according to the eidetic or cognitive laws of consciousness this is specifically relative to Inner you can understand how temporal temporal flow structures how agents are able to make choices act to self-evidence connect the passive structure with the active component of decision making or

goal Pursuit individuals have drives and teleology insofar as they have preferences these preferences are protented goals reach towards by what we would call potentially the Fiat where in the Fiat constitute the willed act towards a desire so the Fulfillment of retention by Fiat drives the agent towards the teleology across a trail set in a retentional extension more information is a matter so the longer your retention more information can be caught in a moment as individuals of mass enough information they can then find a structure to the information which becomes a map rather than a territory and it's computationally more efficient than the sum of all the information amongst this new state of information may lead individuals closer to a shared goal the information of mass turns into an emergent whole of the retention which grabs focus at the present moment because it seems to bring signals to the model that the potential role is closer in its competition is simpler to retain so sharing of retentions and protections among group members any social context can facilitate coordination and the achievement of a shared goal by aligning their beliefs about the past or retentions and expectations about the future or pretensions group members can better predict each other's actions and intentions this allows them to work together more effectively in some cases group members may have retentions and protections that align naturally due to Common experiences or understanding even without active communication however group members often actively communicate their retentions and protentions aligning their beliefs and expectations through dialogue and interaction thereby enhancing their ability to predict each other's actions and interactions emerge jointly within the group not confined to any individual agents unvelt this complex interplay of individual and Collective Dynamics can be explored further in hierarchical generative models encompassing resonant cognitive models communication and even joint emergence the structure of time Consciousness encapsulated in retention and protection thus plays a significant role in the emergence of shared potentials as agents retain and process similar elements from their environment they can co-compute share protections by mapping each other's cognitive Pathways and building upon one another's trails this process is formalized by Bayesian updating the relationship between the topology of protensions and the expectation gradient as well as the role of retention in mapping past knowledge on the current beliefs plays a role in development of shared narratives and goals we can use this to get a deeper insight into the processes that drive the emergence of collective Behavior and the development of shared understanding among agents there's no better way to do it than using active inference by extending generative models of individual agents to account for the beliefs and goals of other agents this is the kin2 theory of mind it enables them to anticipate and adapt

to the actions of their peers by offloading to the group we discussed her little bit earlier shared goals potential goals are future oriented objectives that individuals or groups strive to achieve they're not just mere desires but they're actively pursued through the act of will that we called earlier Fiat the Fulfillment of a protention drives the agent towards a teleology that trail set we discussed of a prediction entails less surprised and therefore the agent's self-evidences by seeking to accurately predict their surroundings agents select a series of actions or policies which generate transitions between states that produce expected observations so through the choice of policy an individual can reach their protented goal by minimizing the expected free energy that a given policy affords them this seems pretty Central and simple in terms of active inference so further protections are represented by the policies of longer time depth in a hierarchical model this corresponds to a policy tied to a higher level action in a slower time scale the choice of policies driven by the expected free energy of that higher order policy or the entire multi-scale generative model is based on the agent's preferences allowing an agent to conduct gold directed behaviors across temporal scales so this is where we can cash out um Jeff yoshimi's Trail sets where agents assimilate actions into the anticipated continuity of objects in their surrounding world this is the simulation ensures a smoother understanding of the environment and helps agents adapt effectively near protensions act as the near present contextual proximity and the far pretensions rely on passive synthesis of deontic cues which are sedimented retentions in the environment the preferences of the agents aligned by policies correspond to attracting sets which in the socio-cognitive setting can be instantiated as narratives having shared narrative is a necessary aspect of any exchange between artifacts people institutions that have some kind of attracting set generalized synchrony which we will discuss a little bit later as well so a team working on a project has a potential goal to complete the project successfully the shared narrative might be the project plan that guides their actions it's like everyone in the team has a shared story about what they're trying to achieve and how they're going to do the relationship between current states and future observations is captured within the B Matrix transitions this Matrix represents the probability of transitioning to a new state based on the current state so the B Matrix transition and the a matrix is utilized to determine which new observations is most likely to occur given the current state and potential transitions in the Australian framework the expectation gradient structure plays a crucial role in defining the depth and fulfillment of pretensions the structure represents the varying degrees of certainty and confidence agents have in their predictions about future States the topology thus is inter and the arrangement is interrelation sorry or there is an interrelation

between the protentions or the arrangements of future expectations which is proportional to the expectation gradient the higher the gradient the more detailed and accurate the pretensions are likely to be the potential Horizon represents the static states of the expected futures of the temporal object while the continuation Horizon is The Amalgamated Trail of the temporal object potential and continuation Horizons are carried in protentions while the imminent Horizon consists of the Primal so agents can coordinate through co-construction where agents or individuals interact and collaborate to build this shared understanding and this process is deeply interconnected within a complex web where agents typically coexist with other agents or subjects forming the collective often referred to as multi-aging groups swarm ensembles or sandwich by aligning their beliefs to the signal found within their Niche they can better adapt to the environment and develop a shared understanding of their sharing information with other agents is beneficial as it allows them to trust one another and pull their resources this collaborative approach is preferable to a conflict-driven scenario in which every other agent is perceived as a potential source of risk and surprising by sharing information agents could collectively construct a more accurate representation of their environment leading to more effective decision making and action through these interactions shared narratives and goals can emerge from a co-construction of the world and each agent's generative model so these narratives and goals are the result of dynamic process that in fact can be mathematically modeled through the equations that we introduce in the paper that is that this presentation is based on but that I will present a little bit later they capture the process of mutual learning where agents adapt their generative model based on the information gathered from their interactions with others agents may start off with slightly or greatly Divergent generative model so in order to minimize surprise there will be a natural alignment within a group as agents adapt to the evolving dynamics of their environment and social contexts for example again consider this team working on the project even if you have your own understanding of the model or of the project you'll interact and share information and learn from each other and then adapt the model over time you'll develop a shared understanding of the project and a shared goal for its completion this is an instance of co-construction it's like building a house together where everyone contributes to the final structure the emergence of shared narratives and the learning and recognition of them may be made more efficient through certain mechanisms such as language in an experiment uh Illustrated uh in a paper that um Carl was on um two agents share the same sensorium ask each other's questions I think it's a paper on 20 questions and Emma homes um with a shared narrative entailing a Common Language this

illustrates The Marked efficiency and the minimization of free energy simply by having a shared language a common ground so shared protections are common future anticipations or goals that emerge among groups of individuals through co-construction it allows them to better anticipate each other's actions and intentions and work towards a common goal it's strongly tied to the conscious efforts of individuals to work towards that goal and to effectively create an organizational structures where we're engaged in a conversation in an exchange assuming that this exchange is meant to align our regenerative models so think of for instance the football team again they have a shared pretension to win the match they anticipate each other's moves and adapt their strategies based on the ongoing game everyone in the team has a shared vision of Victory and they're constantly adjusting their actions to make it happen so individuals infer each other's mental states by observing cues which they can attribute to hidden causes they use their own model of or models that they have largely recognized about others this process is known as generalized synchrony entailing Mutual predictability and can be interpreted as a collective minimization of free energy in a group setting individuals create cues in the world the direct other agents attentions towards the same signals their models of the world align by sharing similar goals and similar environments and by reinforcing patterns of sampling reality so we could we could try to represent this through polynomials u and morphisms each agent's interaction with the environment corresponds to amorphism and the collective interaction of all agents can be represented as a polynomial amorphism so if a polynomial p is defined uh morphism between two polynomials p and Q consists of a pair of functions f and g such that F Maps P of 1 to Q of one and G Maps P of I to Q of the quantity one this morphism must satisfy the commutative of the requirements meaning that the following diagram commutes T of one a p of I maps to Q the quantity I and $po1$ maps to Q of one so we can see how we could create a framework to further explore the cooperative and competitive Dynamics among agents for multiple agents indexed by I with boundaries p and share environment boundary Q their interaction can represent can be represented by a morphism of polynomials which allows us to understand the shared protection in multi-agent systems and provide the foundation for further exploration of Cooperative Dynamics we allow each agent to predict not only its own behavior but also the behavior of its Companions and the environment's response to their actions so the Dynamics of shared predictions are really useful here and can be properly cashed out mathematically so recursive cognition and prediction on the shared environment of the agent's actions and response to each other's we can use sheeph theoretic and toplless theoretic approaches to

understand multi-agent systems a multi-agent system is a system where multiple individuals or entities interact so for example a traffic system can be considered a multi-agent system or different vehicles or interact but topos is a category that has both spatial and logical structure allowing for the expression of logical propositions and deductions within it so we can use these approaches to construct a coherent understanding of the world from the perspective of multiple agents the internal Universe of Agents which can be seen as mathematical objects containing beliefs perceptions and predictions of individual agents are represented using topoi these internal universes provide a foundation for understanding and representing the complex interactions and shared understanding among multiple agents in this Dynamic environment we try to construct a consensus topos by patching together the internal universes or topoi of multiple agents and the idea is to use the chief theoretic and topos theoretic structures to build a shared so we can thus derive the synchronization of goal oriented actions among agents over time and leverage mathematical insights for category Theory and generalized generative models this connection is the mechanism through which the group synchronizes over time and ultimately achieves their objective agents with the system might possess varying levels of certainty in achieving the objective some may choose to expend less energy temporarily decelerating their progress towards the desired outcome this variation in certainty and approach leads to the emergence of different roles within a group with some agents focusing on computational tasks while others monitor outcomes are engaged in different actions in themselves the nested and distributed structure of generative models in active inference allows for the integration of both short-term and long-term goals agents with similar or compatible preferences can coordinate their actions by aligning their policies and working towards common narratives so this alignment reduces the overall expected free energy of their joint actions increasing the likelihood of achieving shared goals and facilitating and sharing of goals and expectations among group members they can learn which allows them to learn essential role in shaping agents preferences and strategies over time agents can gather new experiences and update their generative model they refine their preferences and methods for achieving goals they can influence the formation and modification of shared narratives by having different perspective over different parts of a goal here the the concept of General Life synchrony is manifested when all agents have learned compatible or harmonized aspects of the overall generative model the synchronization is rooted in the retentional and potential dynamics of agents and can be more accurately described using mathematical models so in conclusion which I just shed light on the concept of shared potential goals and multi-agent

systems we've emphasized the importance of understanding this concept as it plays a crucial in the coordination synchronization of Agents actions and intentions in the future continuously highlight the role of cues in the environment and how they can be simple like marks on the road or more complex like symbols and language the cues become carriers of passive synthesis fire protections and far retention and become wasted minimize free energy as they minimize the amount of any individual has to compute it also allows us to delve into the concept of other evidencing in which the environment is in effect the self and others must learn to navigate in terms navigate in terms of others and self we've provided mathematical models to an equation standardized dynamics of shared protections more accurately in terms of category Theory we've explored the Dynamics of shared retentions and retention in hierarchical generative models which reflect the complex interplay of individuals and Collective Dynamics we've encapsulated the structure of time Consciousness and how that this allows agents to co-compute sharep retentions by mapping each other's cognitive pathways so our approach basically begins an examination of Hospital temporal phenomenology integrated with the hierarchical active inference model polynomial morphisms and home polynomials this enables recursive cognition and prediction of not only an agent's actions but also the environment's response to these actions so I think this comes to the end Daniel I think I'm literally at 30 minutes impeccable thank you Mao yeah great uh presentation awesome topic thank you for joining and looking forward to seeing how the work continues to develop farewell now see you you must come soon see you all right the next session is going to be the final session of this symposium it's been a really Incredible Journey and I think this final session is going to be great as well we have a whole panel here people can use their cameras even though I'm not if they want to welcome everyone all just introduce the facilitator Kurt J mungle so Kurt is a a legend in the active ecosystem and has done incredible work with a variety of researchers and curious individuals in the space so we could think of no one better to host this conversation bring it all together so Kurt I'll pass it to you you'll keep an eye on the live chat anyone can call it if they need any supports otherwise thanks everybody for joining and uh we don't see you yet Kurt but yeah I it would it be all right if I left and then came right back for whatever reason my camera is not working sounds good sounds good okay I'll be right back in about 25 seconds well welcome everyone we will get started in just a second so odd I apologize for that okay well you can hear my voice at least yep just give me a moment address boy okay well we'll do what we can okay so you all can't see me but that's all right so we have this All-Star panel here today which to me means my invitation must have been a

mistake but I'll take it thank you to the active inference Institute and I'll go around this virtual Zoom table and introduce everyone briefly Carl Friston is the welcome principal research fellow and scientific director at the Wellcome Trust Center for Neuroimaging and professor of Neurology and University College London he's also the chief scientist at Versus Anna Lemke is chief of the Stanford Addiction Medical Clinic sorry medicine dual diagnosis clinic at Stanford University her popular books include *Drug Dealer MD* and *Dopamine Nation* Raphael Kaufman is CTO of digital Gaia and the onboard sorry and is on the board of directors of the active inference Institute Bert De Vries is a professor at Eindhoven University of Technology where he directs the bias slab research team and also works with industry is an associate professor in computational Psychiatry of the faculty of medicine at the University of Montreal and the director of the Precision Psychiatry and social physiology laboratory in CHU Saint-Justine in the CHU St Justine Research Center he's also affiliated with Mila or Mila which is Quebec's artificial intelligence Institute and other art science and Consciousness service initiatives my name is Kurt Jamungal and I use them I use my background in mathematical physics to analyze various theories of everything that are proposed these include both the theoretical physics side of grand unification with gravity and dualities other schemes as well as attempting to understand what constitutes Consciousness you can find the podcast by typing in theories of everything onto YouTube or whatever podcast catcher you have so my question to everyone the initial question is what have you been working on in the past few months and what excites you about it we'll start with Bert you seem to be smiling and your name sounds like mine okay so um all right so my name is and uh yeah I lead a lab in uh in an electrical engineering department and uh so our lab is called by flat as you as you mentioned so we are interested in Bayesian inference in in general but more specifically in doing it as fast as possible that that has lots of applications outside active inference also in yeah for signal processing and other control systems but definitely of course also for active inference so have our work over the past few months but even over the past few years has been on developing a toolbox to support or to get as far as we can go with trying to do real-time Bayesian inference or real-time free energy minimization and and trying it in applications um so that's that's the work that I try to lead a team of PhD students who do the real work of course I walk around with a cup of coffee but uh that's what they have been working on great and Professor Lemke please yeah well what I've been working on in the last um I don't know a year or so is spending a lot of time thinking about how it is that a faith framework in particular surrendered to a higher power improves people's lives and I've been really struggling to come up with a way of talking

about this that is inclusive thinking about it talking about it it gets to a kind of core piece of it that I'm interested in which is not so much whether or not God exists and what the proof is or not of whether God exists but really what what is it that changes in people's lives when they undergo a spiritual transformation when they surrender to a higher power why why do their lives get better when they do get better and um I'm really new to the active inference World Daniel Friedman is the one who's introduced me to these ideas I'm really excited about the ways in which the whole active inference model might help me at least think about what's going on there and um so so that's what I'm that's what I'm working on it's a little weird but uh that's what I'm interested in great professor friston you're muted yes that's going to be the title of my next book Carl you're on mute a retro from here possibly the future um I I generally work on what I'm told to work on so I'm just been trying to list what I've been working on um in the past few months interestingly um it actually starts with um Daniel Friedman and his um fascination with um along with Axel constant for evolutionary explanations and speak to him in the scale of three issues that we've been hearing about um in the prior session and so that's what I was working on really a scale-free approach to understanding temporary nested processes as free energy minimizing processes as a kind of active inference and how that would lead to a variational synthesis of evolution and then I was told to work on belief sharing by Mao and colleagues versus um and um worked on that in the context of synthetic language and try to establish a really simple proof of principle that communication was an emergent property of any free Ensemble free energy minimizing system um and the just Open brackets is to make the observation it struck me time and time again uh during the session today how important communication is um you know whether we're talking about birds passing messages or responding to uh you know you've got a new posterior or we're talking about the kind of communication um that Mal was talking about Well everybody's been talking about it doesn't seem to be a really uh and indeed the guy the guy um notion of over connectivity and too much communication or the wrong kind of um it does strike me that's quite Central uh to a lot to the both the theme of the um the workshop um and um the specific presentations that we've heard um and then uh in the past few weeks working on stuff I think that Bert would be um more interested in which is um fast and Frugal um message passing schemes that um enable the evaluation the expected free energy not to suffer from uh the limitations imposed by Deep tree searches so um possibly uh Ashwin has spoke about this earlier on but um you're using ideas from dynamic programming backwards induction to try and um take the pressure off the message passing to get fast and efficient uh evaluations of the expected free energy in sort of

um sort of single unit uh sort of single agent uh active inference what have I been doing this week can't remember no that's what I've been great Raphael Kaufman please just a moment and can everyone else hear what about now yeah great yeah thanks uh and it's contagious I think uh no so I've also been drinking a lot of coffee as I've been caring for my now nine month old daughter so uh congratulations thank you so aside from Steve deprivation what I've been doing is um uh I told you guys about this in the in the session earlier today but uh in a nutshell been working on what we call the the the guy protocol uh or the guy Network which is this uh this um open approach for for building a uh common uh common engine you can think of it as a common runtime and uh in a common uh language for with the building blocks of of shared decision making um in the context of on the Mad crisis and getting to getting to a set of of building blocks for decision making that don't kill us all so that's that's kind of it Rafael how many how much coffee is a lot of coffee well it's espresso so I don't think it's quite comparable to like American coffee and Guillaume Dumas please yes sure can you hear me okay uh so in the context of action inference uh I have mostly focused on uh well the two big topic of my lab so uh Precision Psychiatry and social physiology so on the Precision Psychiatry side uh presented earlier in this Symposium uh work on trying to model theory of mind to being able to evaluate through digital phenotyping the level of sophistication of interaction in patients with neurodevelopmental disorders and that connect with also our interest in the social mind and how we can connect our understanding of how we deal with other people but with also ourselves and so that connect with other work at the Mila with the Deep learning architecture with in order functioning humans uh where we're trying to combine different theories of Consciousness such as Global law workspace and attention schema Theory and in the context of attention schema Theory there is this kind of recycling of self and other mechanisms so that's big work and on the other side I'm very interested also in uh military agent systems and so with the Natalie Castle we're going to join us uh soon we've been working on creativity and the emergence of cultural norms in meditation systems with the idea of applying that to climate action and also a new project that is starting here with the an frf of a big Canadian project on indigenous knowledge and how we can think about new narrative around AI and typically a less solipsistic individualistic way of dealing with AI so that's the two main uh field of research can you explain what you mean by less solipsistic lessons oh geez Louise the Californian view of AI because the narrative is mainly driven by uh us and California right now tends to circle around optimization profits I mean even open AI states that AGI is about optimizing profitable human tasks they put like a profitable in their

definitions in the case of indigenous systems of knowledge we are more talking about Community sustainability and the view already only in the application is less individualistic but also in the cognitive science point of view we we have like the the outcore computationalism that would say the brain is just like a computer in a very restrictive sense and so AI like typically the those Transformers that are massively used right now are kind of like bringing that in a very silicon sense while we are an embodied systems that are constituted by our interaction with others and so like in a way how we can think about artificial intelligence in this kind of more co-constitutive way and through a Developmental and cultural lens mm-hmm okay now speaking of the brain as a computer we frequently frequently hear in these circles the brain as a predictive machine so where does this the quote-unquote brain as a predictive machine have its limits and also is that to be interpreted as implying conscious experience is a predictive machine as well okay and if not why not so I think Carl you'd be a great person to start this off and then we'll go around the table again yeah I'm just mindful of Mao's presentation the past half hour um and a notion of temporal thickness um and I'm sure that's got a lot to do with the necessary conditions for the kind of Consciousness and guessing you're referring to you know that freedom from the moment um and if one reads prediction in its sort of psychological or pretensive or pretension um um nature in terms of being able to predict what will happen in the future I think that the um that's that that's going to be a sort of uh a key bright line between things that um do not possess certain kind of um bentians and things that do um and that bright line just rests basically upon um well from perspective of active infants having a generative model that includes the consequences of your own actions in the future so just by having consequences you're now talking about um the future and the cause the consequences um in the future now become random variables therefore you have to infer them which leads you directly to the notion of planning as inference which means that bright lie is just the difference between things that plan and do not plan um I would guess that that's where you probably want to start in terms of foregrounding the role of prediction um as being an aspect of self-organization and its sort of reading under active inference that characterizes just things from non-conscious things namely the ability to plan does that make sense I have follow-up questions we'll get to them at some point but for now Anna Do you have do you have any comments on that and and Bert as well afterward um well you know I'm I'm new to this field so I'm just familiarizing myself even with the language but I I can respond to of the question the brain is a predictive machine in terms of the work that I do clinically in the kind of Psychopathology that I see when I'm

working with patients um you know we I work primarily with narrative the stories that they tell about their lives and one of the recurring themes I've seen through my work is that when patients tell stories in which they are perpetually the victims of other people's actions of the world they tend to then that tends to be a predictive model for them so that they will then go out into the world and unconsciously create scenarios which will perpetuate their victimhood and that part of getting into Wellness is to stop seeing themselves as entirely the victim of other people or circumstance instead begin to appreciate what they contribute to their life problems so I guess when I think of the limits of the brain as a predictive machine I think in some ways one of the big limits is that it's a very powerful predictive machine that actually allows us to subsequently shape what actually happens to us and or our perceptions of what happens to us which then can perpetuate a false narrative and I'll just give one small example from my own life I've had um you know my conflicts with my mother and one of my beefs about her is that she's a very poor communicator and that whenever I email her I either get a cryptic response or no response at all and it drives me crazy and about five years ago she sent an email asked me some questions I wrote her back and responded and it was clear to me that that required yet a response from her which I never got and that then perpetuated my narrative about her as a very poor communicator and many other negative things and then about three months after I sent that email I found the email in my draft inbox in my draft box so I had never actually responded to through his email I hadn't actually sent the email and that was for me um just a you know a personal uh wonderful example of the ways in which we can our actions can actually be manipulated to support our models and perpetuate falsehoods about the world that we live in right and I I have a quick question Bert just for Anna before we get to you so use the word narrative there how are you defining the word narrative is it the same as model is it a sequence of events like what is the specific definition of narrative you know I never really thought about it in those terms but when I talk about narrative I'm talking about the stories that people tell about their lives because that's sort of my data um and and also about models because what I've discovered about self-narrative is it's not just a way to organize the past it also becomes a road map for the future that's the language that I use but I see it Maps very nicely onto ural's language of modeling the world I see Professor DeVries please um yeah it's it's clear that uh the ability to predict is is the essence of talented decision making um yeah I have a different background right I'm not a psychiatrist so I think about these things in different really the thing that that I think about when I think about prediction is I I would assume that it brain predicts far ahead but things get less accurate right if I predict

that I want to go three quarters around around about I don't care about the centimeter where my car goes when I'm around here I just want to get in in the right lane um and I would assume that the brain doesn't take much it takes less computations if you care about things less precisely but that's very hard in a computer we have this is what uh kind of kills Us in uh in in I think in our current uh way of implementing active inference when we want to predict a deep hat we don't care about the accuracy but we don't know how to do it much cheaper when things are less accurate and so that's still a that's a problem that we that need to be working on um it might be in it might be a key to building or to scaling up active Entrance Agency if we compute messages that we want to know that we don't care about if they are less precise that we also spend much less computational there thank you now Professor Kaufman uh I am not a professor but I'll answer anyway um can you just call me Rafa uh yeah so uh there's uh so many really interesting Avenues here and it's not often that I get the the the pleasure of discussing this kind of stuff with this uh kind of diverse crowd but I do I mean I've been interested in uh in these questions uh for for a pretty long time and I I think what comes to mind is like how active inference has a lens uh for instance it enables us to to to to get another sense of what non-dualist views on Consciousness uh or saying when they when they uh get us experientially to to notice the difference between what we actually perceive and what between the different various different uh uh processes or things that are going on in our in our head and our tendency to like want them all together into the same okay this is this is my this is my experience that that I'm immersed in um so noticing being able to to to uh notice the difference between the the the the the the process of perceiving an act and and acting and that's what it's automated and the narrative where the various narratives that are going on uh uh in parallel to my head whether I'm aware of them or not and uh and our ability to to to just put kind of seemingly arbitrary manner levels on top of it to to try to make sense of it all into to force uh for static experience of having multiple Americans and multiple framings going on at the same time makes uh make sense of that experience you know in a in a way that aligns with our physical positions about about how things are so um I I think that's super interesting and also I also I like that uh like this this view of Consciousness as being kind of uh defined by having a narrative or an internal model that's about self or that at least uh it's about self and that leads us as Carl said to the blindness inference uh that's actually like super deflationary in a way that that people are not uh not used to thinking about it and uh and so it's an exciting it's opening the doors all the sort of exciting interdisciplinary conversations we had on the basis of at least I feel like better uh less talking past each other than we've ever had so I'm excited about that

Aaron how is it that you're using the word deflationary there Carl I know but I asked you about that when we had our podcast together but Raphael a similar question so you said deflationary view when we have narratives of self do you mean it leads us to something that Consciousness is much more than sorry much yeah yeah I mean it's uh not necessarily much less than in the in the uh subjective sense but it's much much less than in the in the scientific sense so uh one example of like one not saying exactly that but but maybe something close to what uh Daniel Bennett calls heterophenology which is basically the the statement that the the sum of what one can say scientifically about Consciousness is is um is equal to what can be can be uh what can be studied and uh and modeled and theorized and communicated and learn and uh agreed upon on the basis of objective data about what people including you but they're also other people say uh about their experience which and of course like what can be measured up neural correlates and whatever which is not the same as are not necessarily the same as our experience so our direct what we think is our direct experience of being conscious content right so one way to look at it is that uh uh is that the science of Consciousness doesn't necessarily have to put the privacy on on people the purpose people think they have qualia uh that that we thought and that tends to this deflationary right it's um yeah you have to explain why people believe they have quality or not white people have quality because it's not it's not necessarily a scientific truth that people have quality right Professor Dumas uh yeah so well on the limits of Brands break you mentioned well which we should be always careful to not move from uh one uh map territory fallacy to another ones for sure so we need to be uh Skeptics and avoid to refine the methods and well the this world Symposium show our this metaphor is super useful and and fruitful but I can see like two main uh limits or at least things where we should be careful so uh following what uh Professor Lemke said like in Psychiatry I think the the looping effects and the way people picture themselves uh can be uh very unpredictable and so the the way uh we think about the brain as a predictive machine I think prediction for patients would have different signification of what we mean by predictive machines in the context of factory inference and in general also like uh the I I can see that certain uh psychologists or anthropologists would have a big Appeal on a sudden for uh active inference and preenergetic principles but taking the the words uh as uh directly what they think the word mean where it needs to be we need to be careful the way to communicate it and typically Following also what Professor debris said about uh predict prediction as being super important for decision making I think we should be careful about the the weirdness of uh cognitive science and how it doesn't necessarily expand to uh non-western educated

industrialized country where maybe the cultural value is not about optimization of your decision making or your profit and so that's the first thing where I think we should be careful it's more like a matter of how to communicate the theory in that case not necessarily a limit of the theory itself the the second one is that the equivalence or not with other Frameworks so like uh I can see uh we are the Mao uh talking about category Theory and I'm very curious about right now among the different Frameworks that are out there how can we Define what is equivalent or not with actually inference to be able to to see those limits uh maybe we are leaving what happened with quantum mechanics in the in the early uh 20th century with different interpretation with different tools to model quantum mechanics and air with artificial intelligence and cognitive Neuroscience we have also all those Frameworks and I think the limitation would be then to not take this interpretation as the only one but uh try to have a crosstalk and a differential diagnosis of which one is the best for explaining what okay now my role is the moderator but I would like you all to to speak to one another so I'm going to ask a question that will serve us well you each will speak on it but as the other person is speaking as sorry as the other people are speaking just think okay is there a question that I have or is there a comment that I have so the question is what are some of the recent advancements or breakthroughs in your respective fields that you find particularly promising and well recent let's say 2020 until now so we'll start off with Raphael you're smiling and you look like a champion oh I was thinking what is my field right uh we're very much in the in the um in the last mile of applying the the wonderful stuff that y'all come up with and uh and uh making it uh useful on the ground but so I'm I'm extraordinarily excited about research like what uh what Bert is doing and uh uh I feel like as uh as you almost saying we're also starting to see some some convergence on uh on different um different ways to get to to the same uh at least the same shape of answers and in some cases even to the same um the same uh results so one example of this is uh work that uh um Chris Chris Fields Carl and others have been doing on uh on the quantum uh basically explaining Quantum information Theory uh and uh and how it relates to to to the free energy principle looking at the free energy Principle as the as the classical limit of of quantum information Theory and I don't pretend to to understand all of it but as somebody who has who is coming from a quantum physics background uh just uh uh being alive to see this this kind of conversions happening which as I mean as as uh as Kion said you know we've been at this at this business of like both classical what's what the hell is going on for over a century now and it's nice to to be at a time where again we're we're starting to talk less about each other and that's that's from a periodical perspective from a practical

perspective I think just uh just finding that we have all the building blocks to to uh create uh fast interpretable um reliable and aligned um decision making systems which includes uh AI systems autonomous systems that have all those characteristics and that it turns out that all the things that the reinforcement learning and uh and the neural networks people thought were very hard or impossible are sort of in the realm of what we can do uh looking at things from an active inference lens and vice versa a lot of things that that active influence models have Incorporated so far so it turns out if you if you toss in a neural network approximator here um and um you know some borrow some other you know massively parallel computation techniques uh it also becomes feasible so it's it's converging yeah for me it's what's remarkable is that there's so many domains of of what we thought was exclusively human or would be exclusively human for decades that in just the past couple years robots or computers seem to be just as good if not exceeding us and I don't know if that's promising or worrisome but anyhow Carl please if you don't mind answering the question and then we'll go Anna then Bert and then Dumas well actually just reflecting upon your one of your observations I think it's very difficult to identify one thing uh you know in a sense what is impressive is the diversity of advances and applications um and I just say that because that's what I was thinking you know over the past six hours is listening to your amazing presentation after the presentation and just you know noting how diverse yeah and yet there's this Common Thread this common commitment uh basically say to your curiosity and and using the the tools that naturalize that kind of sense making curious behavior and communication um that inherit from sort of either maths or category Theory or as rough notes now Quantum information Theory but just to pick up on a couple of things which arrive into this conversation so the work with Chris Fields um you know it sounds um lovely and exciting to bring quantum mechanics into um active inference but that's not the move that I think Chris has um really wanting to Champion the move I think is something that we've all been addressing in one way or another which is um leveraging the scale-free aspects of these this principled approach to self-organization and hopefully self-organization to some kind of generalized synchrony um so that that's where the quantum information Theory gets into the game it is scale free and indeed he will go further and say it's completely background for free and everything is constructed it's just uh so I think that that's a lovely move um because once you've gone skill free you then start to ask deep questions about how you coupled one scale to another scale and in a sense ecosystems is just that how do the your the denizens of an ecosystem um how things that constitute how is it co-constructed what is the structure of it these are all questions about how one scale links to

another skill so I think there have been lots of advances in that direction in many many different fronts um and you can read that either in terms of coupling difference of spatial or um yeah spatial scales um but probably more importantly sort of temporal scales and you see that yeah wherever you look um you'll just come back to what do you mean by a narrative it is exactly I think as I said it it's just a a plan is this a story but notice the story has a temporal aspect to it you know I have narratives about being a good person being a good father being a good scientist I also have narratives about you know I want my cup of coffee or I have to go look after that if I don't but Ralph has to go look after the child um so yeah we've all got narratives at very very different time scales and course if you're um you know just come back to Bert's example of you know I'm an autonomous vehicle and I'm sentient and we're five years into the future and I have to drive around the roundabout yeah what Temple scale and what kind of Temple course graining to define the narrative in narratives would be appropriate for that kind of situation the you know the the ensuing planning so um so to my uh so a short answer to your question I think there have been many many advances um they have um I think what they've had in common is basically transcending either different domains but in particular different scales of application I also agree um with the notion well another I think important and pragmatic Advance um is something that Bert mentioned which is democratization of this technology and I think Ralph also into that you know this is the time you start using this for the for the common good so I think you know things like um RX infer and pi mdp and I didn't know about the the Gaia project uh but it sounds as though there's been great advances there as well so this kind of democratization I think is really important this sort of socialization everybody can play and start to sort of um not to talk past each other I think that that's a that's a very important uh a very important advance and did you say the word scale and variance or scale Independence that's a scaling variant um I actually said scale free now I shouldn't have done that um I meant um I I said scale scale free so the the idea that you can apply exactly the same mechanics and literally for example say from birth perspective the same kind of message passing at different space time scales or um at different levels in a hierarchical model so I've actually got a question for bird um in terms of reactive message passing reactive means that you don't have to um prescribe the scheduling but you in in addressing the problems or the issues that are entailed by um having to specify the scheduling of talking or of message passing um you your um you're bound to deal with time and in a scale invariant context or nested with nested scales for example you have to deal with the temp uh the separation of temporal scales so I I think there's going to be a very important generic

question which technically but will have been thinking about furiously for the past few years and I think implicitly we're all going to have to be addressing soon which is the um how do you put that the timing of your messages when you make a move or when you listen to a patient or when you actually pass a message in on a photograph um how does how are you gonna how are we going to be able to put the separation of time scales into the architecture in a way that speaks to this uh the scaling variance you know the gear project for example how do you integrate um live feed from traffic flow um sensors with fluctuations in the climate you know these kinds of this kind of data comes a very very different uh Temple scales and yet has to be assimilated and modeled um you know in a way that is also I think has to pay take you know due courtesy to that separation of temporal scales Bert could you please recapitulate the question for the audience and then begin to answer it yeah the issue is if you if I mean active for instance agents are nested agents and the higher levels supposed to operate on a larger temporal scale but they should they're also working at a lower resolution right if you if you look far ahead you don't care as much about Precision not that the centimeter level when I go around to around about I don't care the centimeter where I land but if I'm but for the next few milliseconds I do care because it may mean the difference between getting in the ditch or not um I don't want to send so the higher level I want to look very deep ahead but I don't want to send every milliseconds a message to go to to look let's say um a minute ahead because so I need to space it out a lot but then I may miss things um so preferably you would just send inaccurate messages but that only works if you actually have a method to Also let's say use less computational power to compute a less accurate message and we're not good at that yet um there I think I mean what we are thinking about is there's a new field or a new field but there's a a field called probabilistic numerics where um we are used in in math to just compute everything very precisely or as precise as possible and and and do not care about how much computation we spend on it um so in probabilistic numerics um I I hope we can leverage this for uh for message computations um I would like to spend let's say proportionally less computational power on on the accuracy of a message One Way possibly would be to consider a message a latent variable that has an uncertainty by itself but I I don't have a a totally clear answer for for Carl because we haven't solved that either um but there there is a problem in let's say what we do on our I mean our computers were so completely different from computers that from let's say from the um from the from the brain that uh yeah we're spending you're spending just too many computations on messages that in the end are very very inaccurate and that's a problem in what we do on our computers currently yeah so I don't know

how we go I mean what you want to do this on the higher level if you you want to look maybe uh um 10 times farther ahead and spend about the same amount of computation on the lower level that's sort of the uh our goal and uh but but we don't have an answer for that either at the moment and and Anna so please feel free to comment on or ask a question to anyone yeah well I mean I you know I don't I don't have uh anything to contribute unfortunately to how computers work but I can tell you that this idea of temporal scales is something that we Face often in our work with patients um for example addicted patients are very focused on short-term rewards and it's it's a fact in fact their ability to control how they feel in the moment that is partially what drives the addiction so when when I try to adopt url's language of minimizing surprise or minimizing free entropy that's one of the things that people are trying to do on a short-term time scale when they become addicted so I have a young woman who's addicted to nitrous oxide which has a very fast onset of intoxication over the order of seconds and a very fast offset and she says that that's exactly what she likes about it because she's controlling it second by second so when we work with patients to get them out of that short temporal Horizon we actually rely more and more on action and having them change something in their lives namely abstain from their drug of choice for long enough to kind of completely reset their brains and to allow them to see this longer temporal because when they're chasing this short control they're actually not able to see themselves in the longer narrative Arc of their lives so that's that was that's what comes to mind for me I don't think it's going to be helpful to people who are trying to build computers but that's that's the kinds of interventions that I make with humans I have a question about that so you mentioned control in the short term and you've also mentioned that you study the positive effects of having a higher power in your life or or Surrender okay I kind of gave the punch line Away by saying surrendering to the higher power where I was going was okay what's the association between seeing yourself in the largest time frame and a higher power and then also is there something that is akin to giving up control when you yeah look farther and farther into the future such a great yeah that's right at the heart of what I'm very interested in because it's a real Paradox right it's this kind of locus of control within ourselves that really in modern culture we we think is a great thing but when that's taken to an extreme and one example of that is addictive behaviors it's very bad for people and for communities and so what can be what can pull people out of that is this kind of surrender to a to a higher power giving up that locus of control locating that locus of control outside themselves not not necessarily like in a theistic sense um one of the things you know that they talk about an alcoholic synonymous for example is um you know you don't have to

believe in God it just doesn't it's just not you you're not you know you're not driving it and so I would be very curious from the perspective of Ural's understanding of active inference and the free energy principle and how the brain works why is it that um sort of embracing our inability to control what happens in our lives can actually be the very source of healing especially embedded in this really kind of over controlled yeah I would even go so far as to say endemically narcissistic culture like I'm really curious I I don't want to take the conversation you know in a direction yeah please please that's a fantastic question so if anyone has a comment on that please well I have some thought uh and I think this applies um supplies both at the personal level and at the at the global level and I think it has to do with the the my quote from earlier uh brought by Edward Tufte that uh if you're falling from a plane yes maybe an altimeter and some instruments uh might be useful but what you really need is a parachute right so we have this um uh we have this uh presupposition that that are whatever like framing we have we have operating in our day-to-day is going to be okay this this is the right Framing and it's gonna it tends to be the this uh this rational scientific uh framing of of okay very very linear uh consequence for for most stuff and it turns out that even like if you inspect or or day-to-day Behavior you know bigger more complicated models are not necessarily better which is where we get uh the success of heuristics under a bounded computation boundary rationality uh and uh if you scale it up to to uh eight billion humans interacted in a in a resources Planet you know a world of possibilities but also of challenges then you just you have to we have to like drastically lower our our uh our bar for yes how much control we have but also even like how much was where does uh information gathering which diminishing returns where does modeling reach uh diminishing returns uh there is a whole literature on uh in the business World about the expected value of information how much you should actually invest but also in science is also uh known as uh optimal experiment design where basically you know acknowledging that uh that you have a limited budget in terms of how much you can act and how much you can probe and how much you can how much time you can spend thinking about stuff and uh I think what we're doing when we feel like burned out or or exhausted from from overthinking uh or in Italy feeling that that okay we've gone too high and we need uh we need to to uh give ourselves a vacation give ourselves some uh some uh some free time here I think that doing this on a principle in a principled way we're just not not just like taking the heuristics and the signals that are that we inherit from from Evolution but actually being able to figure out uh collaboratively and uh and with with some rigor okay this is uh we don't need to know exactly how much how many degrees the world is going to warm uh by 2000 2100 in

order to know that maybe it's a good idea to to start doing something about the amount of of carbon in the atmosphere or whatever right uh and so I think this leads to this this this no knowledge this idea of a real knowledge economy and of things like abstraction as a service how can you actually um how can you actually build in um build in this this kind of like sophisticated and uh and um sophisticated translation layers that take some of this burden from from ourselves as individuals and even as organizations right uh and just put it out in the world as as value-added Services which is what they are if you have any statements or questions or retorts then please feel free uh yeah thanks uh now I was still uh also uh thinking about the recent breakthrough post 2020 in action inference uh to to me like they were like theoretical uh progress that were very interesting uh we heard already about the the formalism maturity at the mathematical level between military scale and scale free aspect um I I really liked or so the the development of a more multi-agent perspective of active insurance it's very interesting like uh especially like for instance the seeing military agent systems we just heard about like even society as well like as many as one system or many as many subsystems and how we can uh use maybe those formalisms to also deal with policy making that's very interesting venue and uh the emergence of norms culture and ideas are so on because um one thing that I'm still struggling with in in the case of active inference models is like how to get outside the checkerboard uh you you can put a lot in the in the model a priori and uh how you make the model creating new stuff that is not backed inside at the first and I on that like one very uh interesting breakthrough was the application of active inference to morphogenesis I really like the the work that has been done on that and on the technical aspects I think the the two main Focus that I love are all the Deep active inference and how to scale up uh actually in front because it's a strong limitation for the adoption of the the formalism uh if it's not scale enough compared to other framework like deep learning and then um the the link with empirical data I really like for instance the work of Ryan Smith and how to connect with uh electrophysiology and clinical data and I think it's also something very important to Anchor uh the theory in in the real world and have like a fancifiability and empirical validation of those models and what are some of the applications of active inference to morphogenesis so well I'm not the expert Erica would be the the best to answer that but uh I was referring to the the work with Michael Levin and our mother can uh help to create a sort of embryogenesis uh so maybe Carl you can explain better than me please yeah I don't have a reputation for explaining things very clearly but yeah that's absolutely right um it was just at work with Michael Evan and colleagues um uh showing that um you can get quite expressive and bimetic out in formation

and movement of different cells into an organization um often described in terms of morphogenesis um simply by communicating your beliefs um so I'm coming back to this sort of cross-cutting theme of communication so if you just broadcast your beliefs and you're a little fell and you have and there are a little ensemblance of cells and they all have the shared charity model that includes if I was in this position I would sense that now they all have exactly the same generative model the same predictions the same expectations um and they're all um broadcasting their beliefs about where they are and the free energy minimizing solution is just when they're all in a place that they expect that they um receive signals that they would expect to to um receive when they're in this place and of course if that is the same for everybody there's only one Arrangement where each cell Finds Its place so basically it's just knowing your place is is an immersion property of making the world mutually predictable through communication so you know a deflationary account of um but the game I thought you know one of your units as Precision Psychiatry so I thought you're going to talk about Precision so I'm going to do a geom now I'm going to he he talked about morphogenesis I'm going to talk about Precision now I think there's there's a that's a really nice way just to pick up on themes which uh everybody's just mentioned and in particular the pathology of precision uh and by Precision you can reposition uh you know in the sense that Bert was talking about in terms of do you use an unsigned integer or a double you know how much can I coarse grain to my numeric representation or you can use it in terms of course grading in the sense of the renormalization group you know it's just chunking things in hours or years as opposed to milliseconds and minutes or you can look at it in terms of a statistician describing the reliability or the inverse variance of a signal because we have to estimate that um when I say we I mean uh your agents and statisticians and and act accordingly and certainly um in the active in the work of Ryan Smith uh on addiction much of the sort of mathematical explanation for these addictive locked in OCD like a phenomena rests upon a failure to get that course grading that Precision estimation right the reason I I I wonder whether it be just worthwhile revisiting um addiction and psychopathology of its only pathology of Behavior Uh from the point of view of getting Precision wrong and certainly assigning too much Precision to low level processing which is what Bert wants to avoid doing um it distracts me that that kind of story also has mileage in terms of why we are in a state of paralysis when it comes to climate change because I also notice gear that you talked about climate action I've never heard that before but that seems to me to be the important thing why isn't there any climate action and Anna you know what it'll be like you going to the clinic and finding some of the

Parkinson's disease why isn't this person moving you know and interestingly the the computational explanation for Parkinson's Disease is assigning too much Precision to the evidence that you're not moving before you move so you know if you don't if you can't ignore the fact that nothing is changing or your pride beliefs your predictions that I'm going to stand up are going to initiate walking don't get a look at because they are immediately canceled because they've assigned too much Precision to the lower level processing and I'm just wondering whether that that's something that kind of pathology is exactly what the the you know what what Ralph Ralph is trying to reverse by having you know models that hand recommendations at hand that provide a more coarse grained view of things the deeper view Anna would you like to comment on that yeah well I just I love I love this idea of the pathology of precision because I think it it manifests in so many different ways um you know not just among my patient population but I think it's almost like a like a cultural sickness in a way um you know the ways in which we we seem like obsessed with certain types of data and we're missing the big picture so I'm gonna I'm gonna think more about that I'm gonna read more about it I really appreciate the discussions interesting for me and Adam when you're referring to the Pathology of precision do you mean so in a more conscious sense that we're over evaluating something that we don't need to whereas Carl you're referring to it in an unconscious sense like the brain putting too much Precision on something because in Parkinson's it's it's not like you consciously are putting Precision in a place well I think you could look at it sort of like in both cases when I think about addiction um you know people aren't doing it consciously it's that they really do um see this as adaptive and healthy and also even on some levels they can't do otherwise and they're not able to see the true impact of their drug use on their lives they're genuinely not able to see the negative consequences I mean that's that's what contributes to getting caught in that vortex but I mean you could also see it as part what's happened culturally like for example like the whole Wellness industrial complex the way that we now count ourselves um you know through all these different devices and if we could just count our breathing and count our heart rate and take supplements you know then we would somehow reach some levitating state of precise wellness and I don't know I mean you know just kind of this is all this is all kind of new new ideas for me that's interesting so you believe that we can be inundated with Health Data and that that's detrimental to us oh absolutely I see that all the time uh-huh so for me I used to have uh okay maybe I'll take this out of the well this is going out live okay let me be let me figure out how to say this diplomatic diplomatically I used to have a device that would measure my heart rate let's say

that and and my sleep and instead of improving my sleep it led to me becoming obsessed with it and then noticing oh I didn't sleep well I must not be feeling good today as well because apparently there's a high connection between How You Sleep Quality exactly it plus added to that the other layer that I should be able to control it right so with all of this data and information I should be able to I don't know reduce free entropy or whatever how to reduce surprise as you guys talk um you know and I think that's obviously that only goes so far and then can actually contribute to our misery because where why aren't we all you know levitating like the Buddha or whatever um you know when we have all these tools and we can pay attention to all this data so I see this also with people who have productivity tools so and not only that but mechanical keyboards let's say some the reason why I don't have a mechanical keyboard even though I think I'd love it is because I know that why the heck do I care about the clicking sound of a keyboard but if I got one then I'd be like well what's the difference between clicking sound A versus clicking sound B versus clicking sound C and I become obsessed with the trappings of productivity that is sharpening so-called sharpening the ax rather than cutting down the tree there's this phrase that is apocryphal and it said it's attributed to I think Lincoln which is that he'd spend 80 of his time or half the time sharpening the knife than cutting the tree and when you just if you just think and then this is just echoed in productivity circles but it just it can't be the case why would you spend so much time sharpening your ex like look at anyone who cuts down a tree with an X most of the time they're not doing that anyhow so if anyone has any comments on on what was just said please can can I uh just talk about what I find an exciting in my Fields about access sure sure yes um I read it was a long time ago but there was a paper um and it's about that it says well active inference is not really a scientific Loop because it's biased and um and I read that and the the sound of the paper was kind of sort of just so it's not good but I think you know active entrance is is it's maybe not a science book it's an engineering Loop because there's bias and we need buyers in engineering we need to make stuff we need to build stuff we need to have a biased so it's an engineering design cycle I see that everywhere around me um and active inference could be um a complete breakthrough in in engineering right um the fields around me are signal processing I myself in a signal processing uh the department or group and everybody builds algorithms for us I mean an active interest agents it's influence of States then the floor below me they build Control Systems well it's influence over actions then other people are working on machine learning it's influence over parameters active infants could be and you'll like this curve it's a very deflationary view on uh on engineering because everything

is just inference and so rather than building algorithms everywhere if we become really good at implementing minimization we will we will be able to build a great engineering design cycles and and we'll be engineering better machines for us or yeah you know for medical procedures or for other things that are important so it has a tremendous uh application potential in engineering um in in engineering I think in most many fields people have sort of drifted in different directions control theorists have I mean they do almost the same thing as signal processing people but they speak a different language now and you know so uh and everybody and and Signal processing people it's a completely different group from the machine learning people but it's all information processing and this brings it uh is filled can bring it together uh so I think it's the if but the thing is that in order to to make it successful in engineering we need to build an application that uh well that impresses right that it's not like a tic-tac-toe thing that it really needs to impress people it's to be better than you know there's some other control systems but once we do that I think there's tremendous uh application potential because there haven't been enormous breakthroughs in Signal processing and control the last big breakthrough I think it was common filtering and this was 1960s and and I mean it's it's kind of uh it's kind of funny that the essence of what we do in active interest is also carbon filtering so um that's uh yeah I think there's tremendous opportunities for for what we do here for uh for engineering so that's why that's why it's exciting to me and do you mind expanding on what you said about acting and therapists should be helping their patients with that oh uh sure just so I mean one of my critiques of uh you know mental health treatment today is that there's not enough there's not enough encouragement of patients to actually go and act differently in the world as a way of gathering data instead of it often ends up being kind of this World building between therapists and patients not not necessarily ultimately adaptive in the world so I was really just kind of responding to what rap was saying that we need to act in the world I think that's more true now um in you know modern Rich Nations than ever before because we are so incredibly sedentary and interactive of course we're interacting with a virtual world and that's you know good and bad but I mean I think we need to be actually acting in the and so there's there's different forms of therapy as you know there's talk therapy and then there's also cognitive behavioral therapy or Psychotherapy and talk therapy but but cognitive behavioral therapy as far as I understand focuses on on the actions is that incorrect or I mean for you know again treating addiction like you're you're not going to really get that far with cognitive behavioral therapy or anything that's focused on just emotions and cognitions people have to go out and actually um try stop using their stopping their

substance or their addictive behaviors and gather data from that experience and then come back and process it so Anna in your field and this question will go to everyone but in your field and what you study where is the largest gap that you would like to see closed well I mean we're facing a huge Mental Health crisis now we have more and more young people coming in with depression anxiety suicidality addictions of all sorts and these are not necessarily people who are struggling by virtue of trauma or socioeconomic disparity these are people who have really privileged lives in many instances so it's uh it's really a puzzle you know what what is going on for people and I think a big part of it is the fact that people are not having embodied experiences they're not having experiences in the world um and also the experiences that they are having are you know these kinds of very quick fixes um fast Pleasures so are you noticing I think you know the the co-created sort of models through healthier communication that allow people to feel part of a community and also to have like truthful co-created narratives trying to use the language I think that's really important so for example one of the things mentioned once he's excited about one of the things I'm excited about in the field of addiction medicine is mutual support and the proliferation of things like Alcoholics Anonymous but also other mutual help groups a lot of them existing now online and the way that people are together creating healthier narratives enacted together to counteract a lot of the unhealthy narratives that I think are driving a lot of decision making today I wanted to know is there a correlation between the rise in mental health or sorry mental illness or mental health issues whatever we want to call it and at a certain trait of people like is it affecting every is it affecting the population the same so the whole population has raised 20 in terms of how many mental health issues they have per year or is it affecting people who deal with abstractions more and more so for instance we're talking over zoom and some people study abstractions just like us and then there's some people who whose work it is to to do something physical like running or or swim like is it affecting everyone equally or are you noticing that there's some broad trend well the broad trends that are out there are just correlational but the more time that people will spend in the virtual world the more likely they are to suffer from depression anxiety and other mental health problems um you know people haven't really been able to narrow that down to specific content but they have been able to save just the sheer amount of time that spending that you're spending online increases your risk for certain mental health you know poor mental health outcomes okay now Carl if well again if anyone has any comments or questions please just raise your physical hand I can see that and okay Ralph Raphael sorry oh I was just gonna say that uh I think another uh notable trend is that and I just

saw somebody see this on YouTube yet just yesterday that uh young people or disproportionately affected by things like climate grief because they're the ones that are going to be alive to deal with it um and uh I think that applies more generally that I know Peter Sandy already like 30 years or something ago uh wrote about the inescapable network of neutrality the reality that that uh that what we're what we do affects each other right and we built we we took advantage of this huge resource buffer that's called uh the biosphere and uh and uh and Earth to to pretend that it didn't for um for quite a long time and you know got a lot of mileage out of it but it's uh now we're at a point where we know that there's there's a uh a whole generation of people that are coming to grip so the fact that um I'm gonna stop myself from seeing a swear word but oh my God we need to uh we actually need to change everything everything about everything that we do and and we need to do it fast and uh and by the way that it's not just uh what we do in uh it's not just what we do out there it's also what we do inside how we get ready for for how to show up for Life internally right so no wonder that that impacts of my my myself I've dealt with anxiety and uh and uh and a lot of of other things we've had conversations about what what are we doing bringing a daughter into this world and all these kinds of things I think it's it's only natural that that it's it's coming to a head in this way right now you know when you have your hand up and I can't see you um yeah it's connect with the what has just been said um so and your initial question about the the gaps that needs to be closed I think like uh the scale free model of health and mental health particularly would be great like uh we are in silos in biomedical research and uh the fact that someone is uh having depression can come from uh interacting genes as much as interacting people and uh also is related to climate change and so on so how we can have a new health systems that doesn't deal with those silos and integrate those different skills to me it's like really a big challenge but a challenge that Quran models and work that we can see going that direction and I'm very enthusiastic about that Bert yeah the um well I in in my field in engineering active influence is not understood because almost all papers are written by neuroscientists and they're hard to read so I was really happy to hear today but I think it's sanjeeb Nam Joshi who is writing an engineering book on active inference so that will I think it will really help um that together with the availability of good toolboxes for implementing active inference should make a lot of Engineers uh much more enthusiastic about active influence because that's uh it's not something that is and it's not understood at the moment in engineering circles so I hope that the book will be good because I'm and Carl where where are some gaps in the in your research that you'd like to see addressed a little empty space out there yet to be um explored

um but in terms of in in terms of what seems to be emerging from um the session and specifically the past few um answers it it does seem to be um important to have um this very generic just to take births um sort of line that this is just one deflationary simple and probably the right way to understand stuff and to make recommendations or to um describe uh people's actions possibly to themselves in a therapeutic uh context and and as such it should be push button technology and it should be democratized and socialized and um I I think that's the challenge practically um and what one may ask why would you want to do that um for me that there are two um clear imperatives um one is um very abstract and it's not really my uh um within my comfort zone and the other one is in my cumbersome another one that's outside my comfort zone as um is um this notion of interactivity and Hyper connectivity in them and and The Meta crisis that we we heard about um and um and our game also refer to this in terms of what he was trying he was trying to distinguish between a Californian notion of optimality and another kind another way forward um and to me it's it's it's you know a stark contrast with growth is good versus sustainability and of course the maths of the free energy principle is just about sustainability it's just a description of the physics of systems random dynamical systems that self-organized to some non-equilibrium steady state and that is what we are um so you know for me there's something deeply if you like apt about the free energy principle and its coronal risk such as active inference in application to ecosystems and lived ecosystems and realized ecosystems so if that if those basic principles can be brought back um into globalization uh into the you know into the market into fintech into um social media into politics into your climate action I think that would be a good thing um I'm just mindful of you know this struck me in a number of the presentations today um you remember before Bert was saying if you look at the brain which is a really lovely example of a self-organizing system to a non-equilibrium steady state um then it's empty and what did he mean by that it didn't mean you were empty-headed what he meant was it's incredibly sparsely connected now that tells you immediately that a pathology of connectivity is hyper connective to overconnectivity which immediately um well it made me very alert to um the presentation of the meta crisis that one of the first three things that was and underwrote them the the metacris or you know the current crisis were conduct ending with is a destruction of that sparse Delica connectivity that defines thingness and defines ensembles of things technically in terms of Markov blankets so if we want a world in which lots of different things can coexist in some kind of generalized synchrony in a sustainable way new deep sparse connectivity and the pathology the thing that will destroy that is over connectivity um so it seems to be very important that we get that into

uh into play in terms of uh machine learning um artificial intelligence politics fintech climate change and the only way it's going to get there is is epistemically by uh equipping people to um build their own little models and ask their own questions you can't tell people this they've got to learn and they're going to learn it for themselves uh just very quickly because I'm sure yeah we've only got a couple of minutes left um the the the um the other agenda which I'm more familiar with is exactly Anderson and guillaume's agenda which is uh making this work in the context of uh um of Neurology and Psychiatry um so if you can democratize um and socialize this way of describing things so that people can now build models of their particular patient in the other use of precision Psychiatry um that are suspended games unit was called after um so to personalized medicine that is really personalized in the sense that it yo you actually have your digital twin of the of your behavior and then you've got that you know you optimize your digital twin to become a model of your patient and then you can start to do experiments on that on that model behavioral interventions or even share that model very much in the spirit of CBT with the patient and say look this is you this is what would happen if you went out and did this and this is what would happen if you went out and did um you know that that to to my mind and indeed that was the initial motivation for much of this work was actually to build um observation models of psychiatric conditions to work out both the um the the pharmacological and physiological basis on the disruption of the overly the pathology of precision and message passing on the factor graphs that are our brain even though they are very empty and on the one hand but also get that behavior that that key thing that Anna was talking about the active engagement with the lived world uh into into that model and hence active inference and just to conclude you know um that that that activity that sort of physical engagement that embodiment that sort of um four ease and everything else um I think it's really coming to a head now in terms of people's um you know after the large language model after the church EPT moment people the bounce back uh has been What's missing you know what's not there and of course what is not there is agency and embodied engagement with the world um and that's why I think that's the whole there's still a lot of work to be done you know in bringing artificial intelligence red as active inference to in a way that matters to people who you know your the people who can make a difference which is basically everybody but specifically politicians and doctors and the engineers and the like so you all now have 30 seconds to 60 seconds to speak directly to the audience what is the message that you what closing message do you have you're speaking directly to someone who's listening you're a curious person they're interested in active inference they also want to lead Better Lives

hopefully and and and do something propitious so what message do you have for them Anna we'll start with you gosh um I'm just gonna say what pops into my mind right now is that one of the things I have learned from my patients who are trying to get into recovery from severe addictions is something that they call the set aside prayer where they set aside all of the notions that they have about how the world works and try to be completely open and receptive to information coming into their minds and I think that's just a wonderful a wonderful frame or concept for all of us uh living in the world to periodically just take a moment and take these models and just say you know everything I think I know about the world I'm going to temporarily suspend it and I'm just going to be open and when that happens we can be present and learn in a way that it's not possible when we're just trying to reinforce our models great fantastic and Bert and then we'll go Carl and then Rapha and then Guillaume um well I I just enjoyed today very much I thought there was a there was a little material boost for people from let's say psychology Neuroscience but also from for engineers so if you haven't watched some of the talks go go go look through the schedule because some of the talks were really good I think so I really enjoyed that um and then yeah what should I tell people um go work out do do a lot of sports it's good for you oh yeah sorry I was going to make a joke but I can't because um so I'm going to use my 32nd uh just to thank uh Daniel and his team for for this um you know so if you want something to do um you should go and watch the live um get involved with this ecosystem I haven't seen that paper be presented before but I was really impressed with you know with the sort of the active Institute and its openness and it's your welcoming um attitude and vision like the Smithsonian um so if you want to pursue these ideas get involved and if you uh haven't got time just make sure you attend next year's active inference Institute a celebration but thank you Daniel Raphael yeah so I'll second what uh what Carl said and follow along with uh it's it's an invitation not just uh to participate in the in the active defense Institute but also an invitations participate in uh building this uh this Gaia tractor this uh this new way of of doing things that acknowledges the value of growth uh and also the value of sustainability joins it all together in this thing called regeneration and uh it really is uh it really is a collective effort it's a collective learning effort so um so and this also by the way also applies to to the the panelists as well and I think uh obviously what uh board and Carl are doing it has immediate uh things having that have to do with with uh what we're after but one of the main things that we discussed that we keep discussing is also like this uh the inter intersubjectivity and the The Importance of Being able to operate well as humans together and that connects directly to cognitive science Psychiatry on and uh and yeah so

everybody that wants to to to be engaged and and be a part of building a better world should be thinking about how do I how what am I doing with my as an individual or as a as a employee organization or a researcher or as a leader uh or the family member what am I doing how does it contribute to to this uh this new this new non-steady non non equilibrium set of state so yeah that's that's kind of it I probably blew through the 60 seconds but here it is well uh I would thanks also all of you for the discussion and the organizer for what they are doing indeed like the the work of the actual Insurance Institute is very uh lovable and interesting from open science perspective they are really uh embodying that so big kudos to them and uh well uh where the fridge was saying like to do sport is good for you I'm not very good at sports but uh some say that science is a team sport so at least uh I have a good team perspective when doing science and being kind to each other would be the best advice to to everyone well thank you I also would like to you get you get yours too somebody else has to come in from outside the Markov blanket though right well I wanted to just thank you Daniel thank you Daniel and and Raphael and well and also Carl and Anna and Bert and Yom this was a tremendous amount of fun and well I hope I get to speak to you all individually and for I as usual I have way more questions than we were able to get to thank you all thanks guys thank you thanks everyone farewell panelists you're all welcome back anytime um thanks everybody for watching or re-watching right now we're about to head over to the Discord and hang out and talk a little more if you want to and then stay involved get involved so until next time bye bye-bye bye-bye